6.0 NATURE AND NATURAL PHENOMENA (Non-Biological)

6.1 THE QUALITY OR PROPERTY OF MATTER

-TKL- 'PHYSICAL STATE OF MATTER'		
BSC	**Stem 1:** (to be) a solid, something solid; to be/make solid, solidify	**Stem 2:** (to be) a liquid, something liquid; to be/make liquid, to liquefy **Stem 3:** (to be) a gas, something gaseous; to be/make into a gas, to render into a gaseous state
CTE	(to be) a state of being solid	
CSV	(to be) the physical process(es) by which one solidifies something (e.g., freezing, dehydration, oxidation, etc.)	
OBJ	(to be) that which is solid	

NOTE: for the meaning of the old FORMAL stems previously associated with this root, use the following new root:

-RTKL- 'LESS TYPICAL PHYSICAL STATES OF MATTER'		
BSC	**STEM 1:** (to be) something in a semi-solid, semi-liquid, viscous, state (e.g., mud, something gelatinous, slime, etc.)	**STEM 2:** (to be) a plasma; to be/make into a plasma **STEM 3:** (to be) a Bose-Einstein condensate; to be/make into a Bose-Einstein condensate
CTE	(to be) a state of being semi-solid	
CSV	(to be) the physical process(es) by which one makes something semi-solid	
OBJ	(to be) that which is semi-solid	

-CN- 'SUBSTANCE'		
BSC	**Stem 1:** (to be) a fundamental manifestation (e.g., chunk, drop, flume, bit, piece, etc.) of a material substance + substance itself	**Stem 2:** (to be) an instance/act of utilizing, manipulating, handling, or working with a substance; to handle, manipulate, work with, utilizing a substance **Stem 3:** (to be) the source of a material substance / compound
CTE	(to be) the material substance being manifested	
CSV	(to be) the form of a fundamental manifestation of a material substance (i.e., chunk, drop, flume, bit, piece, etc.	
OBJ	(to be) the party/entity in contact with, confronted with, observing, handling, etc. a material substance	

-ŘB- **'QUALITY, ATTRIBUTE OR PROPERTY OF MATTER'** (use with the SUF, EXD, FLC, PTW, ITY, etc. affixes)		
	STEM 1	**STEM2:** (to be) a decrease in the degree of a particular quality/attribute/property of matter **STEM 3:** (to be) an increase in the degree of a particular quality/attribute/property of matter
BSC	(to be) a degree of a particular quality (= intrinsic/inherent behavioral property) of matter	
CTE	(to be) the intrinsic nature/essence of a particular quality of matter (i.e., the existence of the quality within or as part of an entity regardless of its physical manifestation)	
CSV	(to be) the physical manifestation of a particular quality of matter; to manifest a particular quality of matter	
OBJ	(to be) the object/entity having the particular quality of matter	

The Stems and Specifications of the following roots are modeled on the 'QUALITY OR PROPERTY OF MATTER' root above:

-**BG**- 'FULLNESS/EMPTINESS'

-**DB**- 'SOLIDITY/HOLLOWNESS'

-**DG**- 'WEIGHT/MASS'

-**XB**- 'TENSION (= STRAIN / MECHANICAL STRESS)'

-**XD**- 'TIGHTNESS/SLACK' (= TAUTNESS)

-**ĻG**- 'PERMEABILITY / POROUSNESS'

-**MB**- 'ABSORPTIVITY'

-**MD**- 'REFLECTIVITY'

-**MG**- 'TRANSPARENCY/OPACITY'

-**BD**- 'CONCAVITY / DEPRESSED'

-**GD**- 'CONVEXITY / ROTUNDITY'

-**GB**- 'COMPRESSED / SQUEEZED'

-**ŇB**- 'RESILIENCE/DELICATENESS' (= CAPACITY FOR RECOVERY FROM DAMAGE)

-**ŇD**- 'DURABILITY/FRAGILITY' (= CAPACITY FOR NOT BEING DAMAGED)

-**NG**- 'PRESSURE'

-**LB**- 'STABILITY/INSTABILITY'

-**LD**- 'VOLATILITY/INERTNESS'

-**LG**- 'DURATION/TEMPORARINESS' [i.e., the amount of elapsed time over which an event or state occurs]

-**BB**- 'EVEN/LEVEL' (i.e., parallel in comparison to an external linear or planar reference standard)

-**DD**- 'PERPENDICULAR / UPRIGHT' (relative to gravity)

-**GG**- 'STRAIGHTNESS/LINEARITY RELATIVE TO OBJECT ITSELF'

-ŢH- ‘PLUME / CLOUD OF GAS / SMOKE / STEAM / GASEOUS PHENOMENON’		
BSC	**STEM 1:** (to be) a plume/cloud of smoke from an act/process of combustion (e.g., smoke from a fireplace, from an explosion, from a volcano, from a forest fire, etc.)	**STEM 2:** (to be) a plume/cloud of steam/water vapor [**NOTE:** for a cloud in the sky, see the root -**FTH**-] **STEM 3:** (to be) a plume/cloud of gas other than steam or smoke from combustion (e.g., a gas leak)
CTE	(to be) a state of smokiness; for the air to be filled with smoke	
CSV	(to be) an act/process of combustion which gives rise to a plume of smoke; for there to be smoke due to an act/process of combustion	
OBJ	(to be) the plume of smoke itself	

-TPR- ‘CONSISTENCY/MALLEABILITY/FLEXIBILITY/PLASTICITY/RHEOLOGICAL PHENOMENON’		
BSC	**Stem 1:** (to be) a particular degree of malleability, flexibility, plasticity; to have a certain degree of malleability, flexibility, plasticity	**Stem 2:** (to be) a particular degree of squeezability, compressibility **Stem 3:** (to be) a particular degree of elasticity
CTE	(to be) a state of malleability, flexibility, plasticity; to be malleable, flexible	
CSV	(to be) an act of determining the degree of malleability, flexibility, plasticity of something; to determine the degree of malleability, flexibility, plasticity of something	
OBJ	(to be) that which has a particular consistency in terms of malleability, flexibility, plasticity	

-TPW- ‘SPRINGINESS / BUOYANCY / RECOIL’		
BSC	**Stem 1:** (to be) a particular degree of springiness; to have/manifest a particular degree of springiness	**Stem 2:** (to be) a particular degree of buoyancy; to have/display a particular degree of buoyancy **Stem 3:** (to be) a particular degree of recoil; to have/display a particular degree of recoil
CTE	(to be) a state of springiness; to be springy	
CSV	(to be) an act of determining the degree of springiness	
OBJ	(to be) that which has a particular degree of springiness	

-L̩MW- ‘FRAGILITY / BRITTLENESS / CRUMBLINESS’		
BSC	**Stem 1:** (to be) a particular degree of fragility; to have/manifest a particular degree of flimsiness	**Stem 2:** (to be) a partticular degree of brittleness **Stem 3:** (to be) a particular degree of crumbliness
CTE	(to be) a state of fragility; to be fragile	
CSV	(to be) an act of determining the degree of fragility; to determine the degree of fragility of something	
OBJ	(to be) that which has a particular degree of fragility	

-BGW- ‘LIMPNESS / FLACCIDITY / JIGGLINESS / FLABBINESS’		
BSC	**Stem 1:** (to be) a particular degree of limpness/flaccidity; to have/manifest a particular degree of limpness/flaccidity	**Stem 2:** (to be) a particular degree of jiggliness **Stem 3:** (to be) a particular degree of flabbiness
CTE	(to be) a state of limpness/flaccidity; to be limp/flaccid	
CSV	(to be) an act of determining the degree of limpness/flaccidity; to determine the degree of limpness/flaccidity of something	
OBJ	(to be) that which has a particular degree of limpness/flaccidity	

-L̩PW- ‘PLIANCY / DUCTILITY / SUPPLENESS / DEFORMABILITY / SPREADABILITY’		
BSC	**Stem 1:** (to be) a particular degree of pliancy/ductility/suppleness; to have/manifest a particular degree of pliancy/ductility/suppleness	**Stem 2:** (to be) a particular degree of deformability or “dentability” (i.e., how easily deformed or dented a material is) **Stem 3:** (to be) a particular degree of spreadability
CTE	(to be) a state of pliancy/ductility/suppleness; to be pliant/ductile/supple	
CSV	(to be) an act of determining the degree of limpness/flaccidity; to determine the degree of pliancy/ductility/suppleness of something	
OBJ	(to be) that which has a particular degree of pliancy/ductility/suppleness	

-CKW- ‘FLIMSINESS / COHESIVENESS / STRUCTURAL INTEGRITY / STABILITY’		
BSC	**Stem 1:** (to be) a particular degree of flimsiness; to have/manifest a particular degree of flimsiness	**Stem 2:** (to be) a particular degree of cohesiveness (tendency to not break apart) **Stem 3:** (to be) a particular degree of structural integrity/stability
CTE	(to be) a state of flimsiness; to be flimsy	
CSV	(to be) an act of determining the degree of flimsiness; to determine the degree of flimsiness of something	
OBJ	(to be) that which has a particular degree of flimsiness	

-LLW- ‘LIGHT / ILLUMINATION / BRIGHTNESS’		
BSC	**Stem 1:** (to be) a state of being bright/luminous; to be bright/luminous (i.e., either emanating or reflecting a relatively large amount of light from either external or internal illumination)	**Stem 2:** (to be) a state of being bright/luminous due to internal illumination (i.e., entity is, or carries within, its own source of illumination **Stem 3:** (to be) a state/act of being bright/luminous due to high surface reflectivity
CTE	(to be) the state of being bright/luminous	
CSV	(to be) a physical act of brightening/illuminating; to brighten, to illuminate	
OBJ	(to be) something that is bright, highly reflective, or lit up, (to be) a (source of) light	

-ŘM- ‘DIMNESS / PALENESS / SHADOW’		
BSC	**Stem 1:** (to be) a state of being dim or act of dimming; to be/make dim, to dim (i.e., difficult to see, not very visible due to insufficient external or internal illumination)	**Stem 2:** (to be) a state of being pale/blanched/pallid/wan; to be/make pale/pallid/wan/blanched **Stem 3:** (to be) a state/act of being/placing in the shade or in shadow; to be/place in the shade or in shadow
CTE	(to be) the state of being dim	
CSV	(to be) a physical act of dimming; do something to make an entity dim	
OBJ	(to be) something that is dim	

-ŘMW- ‘DULLNESS / FADEDNESS / MURKINESS / GLOOMINESS’		
BSC	**STEM 1:** (to be) a state/act of being/making something dull/lackluster in appearance; be/make dull/lackluster in appearance (i.e., decreased reflectivity)	**STEM 2:** (to be) a state/act of being something faded; to be faded, to fade (i.e., loss of former degree of brightness/visibility due to age, wear, decay) **STEM 3:** (to be) a state/act of being something murky/gloomy; be/make murky/gloomy
CTE	(to be) the state of being dull/lackluster	
CSV	(to be) a physical act of making dull/lackluster; do something to make an entity dull/lackluster	
OBJ	(to be) something that is dull/lackluster	

-ŽX- ‘FIRE / BURNING / MELTING / CALEFACTION’		
BSC	**Stem 1:** (to be) a state/act of burning with/by fire; to burn by fire	**Stem 2:** (to be) a state of a substance being/becoming heated by an external source of heat; to heat (up), to warm (up) **Stem 3:** (to be) a state of melting/liquefaction; to melt, to liquefy via heat
CTE	(to be) a state of being burned, being on fire; to be on fire, to be in flames	
CSV	(to be) an act of igniting or combustion; to ignite, to combust, to catch fire	
OBJ	(to be) a flame [Derivatives: a fire, a blaze, a conflagration, etc.]	

-KV- ‘COLDNESS / COOLING / FREEZING’		
BSC	**Stem 1:** (to be) a state/act of being/becoming cold/frigid; to be(come) cold/frigid	**Stem 2:** (to be) a state of a substance being/becoming cool/cold by an external source of cold; to cool (down), to get cold **Stem 3:** (to be) a state of being frozen; to freeze
CTE	(to be) a state of coldness, a state of being cold	
CSV	(to be) an act of chilling; to chill something	
OBJ	(to be) a tangible sign/indicator of coldness (e.g., ice, snow, frost, etc.)	

Derivations: air conditioner, refrigerator

-GÇ- ‘DISCARD AS RUBBISH / TRASH / GARBAGE / DETRITUS / REFUSE / RUBBLE / DEBRIS’		
BSC	**Stem 1:** (to be) an act/process of discarding something as rubbish/trash/garbage/refuse	**Stem 2:** (to be) an act/process of discarding something as junk/flotsam/ dross/offal/leavings/scraps/remnants (i.e., worthless leftover material from some productive or consumptive process) **Stem 3:** (to be) an act/process whose material result/outcome constitutes rubble/debris/detritus/remains (i.e., disintegrated remnant material)
CTE	(to be) a state of being/constituting rubbish/trash/garbage/refuse	
CSV	(to be) a physical act of discarding something as rubbish/trash/garbage/refuse	
OBJ	(to be) an object/item/entity/amount of material discarded as rubbish/trash/garbage/ refuse	

-DGL- 'PLANAR ORIENTATIONAL INVERSION'		
BSC	**Stem 1:** (to be) a state/act of being/placing sideways; to (be) position(ed)/orient(ed) sideways (relative to an observer)	**Stem 2:** (to be) a state/act of being/placing upside-down; (be) position(ed)/orient(ed) upside-down **Stem 3:** (to be) a (to be) a state/act of being/placing backwards [i.e., only in sense of facing the opposite direction than expected]; (be) position(ed)/orient(ed) backward [i.e., facing the opposite direction]
CTE	(to be) the state of being in a sideways position	
CSV	(to be) an act of physically orienting/placing something sideways (relative to an observer)	
OBJ	(to be) the entity which is positioned sideways	

-BGL- 'TOPOLOGICAL INVERSION'		
BSC	**Stem 1:** (to be) a state/act of being a mirror-image; to invert in a mirror-image fashion	**Stem 2:** (to be) a state/act of being/placing ; to (be) position(ed)/orient(ed) inside-out **Stem 3:** (to be) a state/act of being a mirror-image both laterally and vertically (i.e., mirror-image plus upside-down)
CTE	(to be) the state of being a mirror-image	
CSV	(to be) an act of physically causing/making a mirror-image	
OBJ	(to be) the entity which is mirror-imaged	

-KSW- 'BIFURCATION / MULTIFURCATION / BRANCHING'		
BSC	**Stem 1:** (to be) a state/act of forking into two, bifurcation; to bifurcate	**Stem 2:** (to be) a state of forking into three or more branches/paths, etc. **Stem 3:** (to be) a state of continous (bi-)furcation/branching, so that the number of branches/paths multiplies over time or space (e.g., tree branches, a philogenic taxonomic chart, etc.)
CTE	(to be) a state of being bifurcate(d)	
CSV	(to be) a physical act/process of bifurcation	
OBJ	(to be) that which forks into two, that which is forked in two	

-L̤R- 'BALANCE / EQUILIBRIUM / PROPORTION'		
BSC	**Stem 1:** (to be) a state/act of being in balance / act of balancing; to balance / to be/make balanced (i.e., so that a state/entity/construction is stable)	**Stem 2:** (to be) a state/act of equilibrium **Stem 3:** (to be) a state/act of a set of entities being in proper proportion to one another; to be in proper proportion to one other (i.e., in terms of optimal functionality or well-being)
CTE	(to be) a state of balance	
CSV	(to be) an act of balancing; adjust the position/state of something so that it is in balance	
OBJ	(to be) that which is balanced or in balance	

-TḶR- ‘SYMMETRY / PARITY / HOMEOSTASIS’		
BSC	**STEM 1:** (to be) a state/act of being symmetrical, or having/displaying symmetry; to be/make symmetrical, to have/display symmetry	**STEM 2:** (to be) a state/act of being in parity; to be/place in parity **STEM 3:** (to be) a state/act of homeostasis; to be/make homeostatic, have/display homeostasis
CTE	(to be) a state of symmetry	
CSV	(to be) an act of making something symmetrical; to make symmetrical	
OBJ	(to be) that which is symmetrical	

-ÇŇ- ‘CURL / COIL / LOOP’		
BSC	**Stem 1:** (to be) a state/act of curling; to curl, to (be)form(ed) into a curl (both the shape/form and the entity so shaped/formed)	**Stem 2:** (to be) a state/act of looping, being in a loop; to (be) form(ed) into a loop **Stem 3:** (to be) a state/act of coiling, being coiled; to coil, be formed into a coil
CTE	(to be) in (the form of) a curl, be curled, be curly	
CSV	(to be) an act of forming into a curl; to (form into a) curl	
OBJ	(to be) that which is curled	

-RÇŇ- ‘TWIST / TANGLE / KNOT’		
BSC	**STEM 1:** (to be) a state/act of twisting; to twist, be twisted, form into a twist [i.e., curled in more than one planar dimension]	**STEM 2:** (to be) an act/state of being knotted, tied in a knot; to knot, tie in a knot **STEM 3:** (to be) a state/act of tangling; to tangle, be tangled, form into a tangle
CTE	(to be) in (the form of) a twist; to be twisted	
CSV	(to be) an act of twisting; to (form into a) twist	
OBJ	(to be) that which is twisted or in a twist	

6.2 MEASUREMENT OF NATURAL PHENOMENA

-JGW- 'MEASUREMENT OF A LINEAR DIMENSIONAL PROPERTY'		
BSC	**Stem 1:** (to be) a state/act of being/having/making something of a certain measure of a linear dimensional property (i.e., length, height, width, circumference, etc.)	**Stem 2:** (to be) a state/act of measuring a linear dimensional property of something; to measure the length or width or height or circumference, etc. of something **Stem 3:** (to be) a particular system of linear dimensional measurement; to utilize a particular system of linear dimensional measurement
CTE	(to be) the state of having/being a certain measure of a linear dimensional property	
CSV	(to be) a physical act of making something have/be a certain measure of a linear dimensional property; to make something a certain number of measurement units [long or wide or high or around, etc.]	
OBJ	(to be) the actual length/height/width/circumference, etc. of something	

The following affixes are available for use with the above root.

-rtn	LDA Linear Dimensional Measurement A *
1	(in) Planck lengths
2	(in) angstroms
3	(in) microns
4	(in) millimeters
5	(in) meters
6	(in) kilometers
7	(in) astronomical units (AUs)
8	(in) light-years
9	(in) parsecs

-rtm	LDB Linear Dimensional Measurement B *
1	(in) inches
2	(in) feet
3	(in) yards
4	(in) fathoms
5	(in) rods
6	(in) furlongs
7	(in) miles
8	(in) nautical miles
9	(in) leagues

-rtk	ARM Areal Measurement
1	(in) barns
2	(in) squares
3	(in) acres
4	(in) roods
5	(in) hectares
6	(in) square perches or sq. rods or poles
7	(in) tetrads
8	(in) townships
9	(in) myriads

* The Type-2 form of these affixes refers to the measurement unit squared, e.g., square meters, sq. parsecs, sq. yards, sq. miles, degrees squared, arc-seconds^2, steradians, etc.

The following roots use the same Specification pattern as the above root:

-**JGY**- 'MEASUREMENT OF PLANAR AREA' as measured via the Type-2 forms of the **LDA/LDB** or **ARM** affixes shown above

-**JGV**- 'MEASUREMENT OF A 3-DIMENSIONAL VOLUME' (as measured via the affixes shown below)

-řw	VMA Volumetric Measurement A
1	(in) cubic Planck lengths
2	(in) cubic angstroms
3	(in) cubic microns
4	(in) cubic millimeters
5	(in) cubic meters
6	(in) cubic kilometers
7	(in) cubic astronomical units (AUs)
8	(in) cubic light-years
9	(in) cubic parsecs

-řy	VMB Volumetric Measurement B
1	(in) cubic inches
2	(in) cubic feet
3	(in) cubic yards
4	(in) cubic fathoms
5	(in) cubic rods
6	(in) cubic furlongs
7	(in) cubic miles
8	(in) cubic nautical miles
9	(in) cubic leagues

-řl	VMC Volumetric Measurement C
1	(in) minims
2	(in) fluid drams
3	(in) teaspoons
4	(in) tablespoons
5	(in) milliliters
6	(in) liters
7	(in) fluid ounces
8	(in) gallons
9	(in) barrels

The following roots use the same Specification pattern as the above root:

-**ÇPH**- 'MEASUREMENT OF A PLANAR ANGLE' as measured via the ANG affix shown below

-**ÇTH**- 'MEASUREMENT OF A SOLID ANGLE' as measured via the Type-2 forms of the ANG affix shown below

-dg	ANG Angular Measurement *
1	(in) points
2	(in) hour angles
3	(in) grads
4	(in) mils
5	(in) radians
6	(in) sextants
7	(in) seconds of arc / (in) arc-seconds
8	(in) minutes of arc / (in) arc-minutes
9	(in) degrees

* The Type-2 form of these affixes refers to the measurement unit squared, e.g., degrees squared, arc-seconds^2, radians^2 (i.e., steradians), etc.

-STR- 'DURATION / TAKING UP A SPAN OF TIME / SPENDING ONE'S TIME'			
	Stem 1	**Stem 2**	**Stem 3**
BSC	(to be) a state/act of being/lasting/enduring making something last a certain amount of time; to be a certain amount of time in duration, to last for a certain amount of time	(to be) a state/act/process of something taking up/requiring a certain amount/span of time; for something to take up/require a certain amount of time, to take up/require a certain amount of time	(to be) a state/act/process of spending a certain amount/span of (one's) time doing something; to spend a certain amount/span of time doing something
CTE	(to be) the state of lasting/being a certain amount of time, a state of temporal duration	(to be) the state of taking up/requiring a certain amount/span of time, a state of required temporal duration	(to be) the state of spending a certain amount/span of (one's) time
CSV	(to be) a physical act of making something last/be/endure a certain amount of time; to make something last a certain amount of time	(to be) a physical act of waiting (for)/enduring something that takes up a certain amount/span of time; to wait for/ endure something taking up/requiring an amount of time	(to be) a physical act of spending a certain amount/span of (one's) time; to go about an activity or activities which constitute a certain expenditure of (one's) time
OBJ	(to be) the actual amount/length of time something lasts/endures/takes, a duration	(to be) the actual amount/length of time something takes/requires, a required duration/span of time	(to be) the actual amount/length of time one spends doing something, the amount/span of one's time expenditure

-KSTR- 'TEMPORAL MEASUREMENT'		
BSC	**Stem 1:** (to be) a state/act of measuring a certain amount/span of time	**Stem 2:** (to be) a state/act of recording/archiving/making available/accessing a temporal measurement made **Stem 3:** (to be) a particular system of timekeeping employed in a temporal measurement (i.e., what system of measurement units are employed)
CTE	(to be) the state of there being a span of time to measure	
CSV	(to be) a physical act of measuring an amount/span of time; to perform a temporal measurement	
OBJ	(to be) the actual amount/length of time measured	

The following roots use the same Specification pattern as the above root:

-**LSTR**- 'MEASUREMENT OF CYCLIC/PERIODIC FREQUENCY' as measured in the number of cycles per unit of time

-**RSTR**- 'MEASUREMENT OF RECURRING BUT NON-CYCLIC/NON-PERIODIC EVENTS' as measured in number of events per unit of time

The following affixes are available for use with the above temporal measurement roots.

-řfs	USM Ultra-Short Temporal Measurement *
1	in/per yocto-second(s) (10^{-24} seconds)
2	in/per zepto-second(s) (10^{-21} seconds)
3	in/per atto-second(s) (10^{-18} seconds)
4	in/per femto-second(s) (10^{-15} seconds)
5	in/per pico-second(s) (10^{-12} seconds)
6	in/per nano-second(s) (10^{-9} seconds)
7	in/per micro-second(s) (10^{-6} seconds)
8	in/per millisecond(s) (10^{-3} seconds)
9	in/per second(s) (e.g., with root -**LSTR**- above = Hertz)

-rns	TPM Temporal Measurement
1	in/per minute(s)
2	in/per hour(s)
3	in/per day(s) (24-hour periods)
4	in/per week(s)
5	in/per month(s)
6	in/per calendar year(s)
7	in/per decade(s)
8	in/per (human) generation(s)
9	in/per (human) lifetime(s)

-rňs	LTM Long-TermTemp. Meas.
1	in centuries/per century
2	in millenia /per millenium
3	in/per 10,000-year period(s)
4	in/per 100,000-year period(s)
5	in/per age(s) (i.e., 1 million years)
6	in/per epoch(s) (i.e., 10 million years)
7	in/per era(s) (i.e., 100 million years)
8	in/per eon(s) (i.e. 500 million years)
9	in billions of years /per billion years

* this affix is the measurement counterpart to the **UST** affix

The following roots follow the same Specification pattern as the previous root and utilize the above affixes as well:

-**LKK**- 'ANGULAR VELOCITY OR ANGULAR FREQUENCY' as measured in units of planar angle measurement per unit of time

-**LKKW**- 'SNAP OR JOUNCE' as measured in distance per unit of time to the fourth power

-**LKKY**- 'JERK, JOLT, SURGE OR LURCH' as measured in distance per unit of time cubed

-**LKKL**- 'YANK' as measured by mass times distance/time^3

-**LKKR**- 'KINEMATIC VISCOSITY OR DIFFUSIVITY COEFFICIENT' as measured in distance squared per unit of time

-**LKKŘ**- 'VOLUMETRIC FLOW' as measured in distance cubed per unit of time

-**RKK**- 'SPREAD RATE BY VOLUME' as measured in distance cubed per distance squared

-**RKKW**- 'ENERGY DENSITY' as measured by energy per cubic distance

-**RKKY**- 'SURFACE TENSION' as measured by force per distance

-**RKKL**- 'STIFFNESS' as measured by force per distance

-**RKKR**- 'RATE OF ABSORBED DOSE OF IONIZING RADIATION' as measured by energy per mass/time

-**NKKW**- 'SPECIFIC ENERGY' as measured by energy per unit mass

-**NKKY**- 'RADIANT EXPOSURE OF A SURFACE / ENERGY DISTRIBUTION OVER A SURFACE / INSOLATION / SOLAR RADIATION' as measured by energy per square distance

-**NKF**- 'SPECTRAL FLUX BY FREQUENCY' as measured by power per cycle/time

-**NKFW**- ‘SPECTRAL FLUX BY WAVELENGTH’ as measured by power/distance

-**NKFY**- ‘HEAT FLUX DENSITY / IRRADIANCE / RADIANT EXITANCE / RADIOSITY’ as measured by power/distance2

-**NKFL**- ‘SPECTRAL EXITANCE / RADIOSITY / IRRADIANCE BY FREQUENCY / STRENGTH OF RADIO WAVE EMISSION’ as measured by power/distance^2 per cycle/time

-**NKFR**- ‘SPECTRAL EXITANCE / RADIOSITY / IRRADIANCE BY WAVELENGTH’ as measured by power/distance^2 per distance

-**NKFŘ**- ‘RADIANT INTENSITY’ as measured by power/distance^2 per steradian

-**NKFM**- ‘SPECTRAL INTENSITY’ as measured by power/distance^2 per steradian per cycle/time

-**NKFN**- ‘RADIANCE’ as measured by power/distance^2 per steradian per distance^2

-**NKFF**- ‘SPECTRAL RADIANCE BY FREQUENCY’ as measured by power/distance^2 per steradian per distance^2 per cycle/time

-**ŘKK**- ‘DYNAMIC VISCOSITY’ as measured by pressure multiplied by time

-**ŘKKW**- ‘ACOUSTIC IMPEDANCE’ as measured by pressure multiplied by time per distance

The above roots utilize the following affixes:

-ḷp	EMU Energy Measurement Units
1	(in) Planck energy units
2	(in) electronvolts
3	(in) ergs
4	(in) hartrees
5	(in) joules
6	(in) therms
7	(in) calories
8	(in) thermies
9	(in) quads

-ḷt	FMU Force Measurement Units
1	(in) Planck force units
2	(in) dynes
3	(in) poundals
4	(in) pound-force
5	(in) newtons
6	(in) kiliponds
7	(in) sthènes
8	(in) kips
9	(in) ton-forces

-ḷk	PMU Pressure Measurement Units
1	(in) bars
2	(in) baryes
3	(in) pounds per square. inch
4	(in) torrs
5	(in) pascals
6	(in) pièzes
7	(in) millimeters of mercury
8	(in) inches of mercury
9	(in) standard atmospheres

-ḷv	OEM Other Energy Measurement Units
1	(in) watts
2	(in) horsepower
3	(in) sieverts
4	(in) grays
5	(in) rads
6	(in) poises
7	(in) poiseuilles
8	(in) rayls
9	(in) janskys

For ‘VELOCITY’ and ‘ACCELERATION’ use the following root:

-FM- **‘DEGREE OF VELOCITY’** Affix: **VEL** (use SUF/EXN/EXD, etc.)		
BSC	**STEM 1:** (to be/manifest a) degree of speed/velocity [= ratio of distance over time]	**STEM 2:** (to be/manifest a) degree of acceleration (i.e., increasing rate of speed/velocity) **STEM 3:** (to be/manifest a) degree of deceleration (i.e., decreasing rate of speed/velocity)
CTE	(to be) something with a degree of speed/velocity	
CSV	(to be) a degree of speed/velocity	
OBJ	(to be) an entity/party impacted/affected by one’s degree of speed/velocity	

In addition to the **VEL** affix associated with the above root, the following affix provides for exact measurement of velocity and acceleration:

-ng	**VAM Velocity and Acceleration Measurement** [**NOTE:** Use Type-2 affix to indicate time unit squared for measurement of acceleration]
1	kilometers per hour
2	kilometers per second
3	meters per second
4	millimeters per second
5	microns per second
6	miles per second
7	feet per second
8	inches per second
9	miles per hour

-LPS- ‘PHYSICAL MASS’ Associated Affix: **MSS**		
BSC	**Stem 1:** (to be) a state/act of having a certain mass	**Stem 2:** (to be) a state/act of measuring (or observing a measurement of) the mass of something; to measure or view a measurement of the mass of something **Stem 3:** (to be) a particular system of measuring mass; to utilize a particular system of measuring masst
CTE	(to be) the state of having a certain mass	
CSV	(to be) a physical act of making something have a certain mass	
OBJ	(to be) the actual mass of something	

The following affix is available for use with the above root.

-lps	MSS Measurement of Mass
1	(in) Planck masses
2	(in) daltons
3	(in) grains
4	(in) grams
5	(in) kilograms
6	(in) metric tonnes
7	(in) pounds
8	(in) (short) ton (= 2000 lbs.)
9	(in) solar masses

The following roots have the same Specification pattern as the root -**LPS**- above:

-**LPSW**- 'ENERGY/WORK' (as measured in mass times distance squared divided by unit of time squared)

-**LPSM**- 'FORCE' (as measured by mass times distance divided by unit of time squared)

-**LPSN**- 'PRESSURE' (as measured by (mass x (distance/time^2)) / distance^2)

-**LPSF**- 'POWER' (as measured by mass x distance^2 / time^3)

-**RPS**- 'MOMENTUM' (as measured by mass times distance/time)

-**RPSW**- 'ANGULAR MOMENTUM' (as measured by mass times distance^2 /time)

-**RPSL**- 'THRUST' (as measured by mass times distance/time^2)

-**RPSR**- 'TORQUE OR MOMENT' (as measured by mass times distance^2 /time^2)

-**RPSM**- 'LINE DENSITY' (as measured by mass per distance)

-**RPSN**- 'VOLUMETRIC DENSITY' (as measured by mass per volume)

-**ŘPS**- 'SPECIFIC VOLUME' (as measured by volume per mass)

-**ŘPSW**- 'SPREAD RATE' (as measured by mass per area)

-**ŘPSL**- 'AREA DENSITY' (as measured by mass per area)

-PṬK- ‘TEMPERATURE’		
BSC	**Stem 1:** (to be) a state/act of having/being at or causing to have a certain/particular temperature; to be/have/make a certain temperature	**Stem 2:** (to be) a state/act of measuring (or observing a measurement of) the temperature of something; to measure or view a measurement of the temperature of something
CTE	(to be) the state of having/being at a certain/particular temperature	
CSV	(to be) a physical act of cooling/heating something to a certain temperature	**Stem 3:** (to be) a particular system of temperature measurement; to utilize (for measurement or calibration/cooling/heating) a particular system of temperature measurement
OBJ	(to be) the actual temperature of something	

The following affix is available for use with the above root.

-šj	TMS Temperature Measurement System
1	(in) Planck temperature / (in) T_P
2	(in) degrees Newton / (in) °N
3	(in) degrees Celsius / (in) °C
4	(in) degrees Rankine / (in) °R
5	(in) kelvins
6	(in) degrees Delisle / (in) °D
7	(in) degrees Fahrenheit / (in) °F
8	(in) degrees Rømer / (in) °Rø
9	(in) degrees Réaumur / (in) °Ré

The following roots have the same Specification pattern as the root -**PṬK**- above:

-PṬKW- ‘HEAT CAPACITY OR ENTROPY’ (as measured in energy per unit temperature)

-PṬKY- ‘ENTHALPY’ (as measured in units of energy)

-PṬKL- ‘SPECIFIC HEAT CAPACITY OR SPECIFIC ENTROPY’ (as measured in energy per mass times temperature)

-PṬKR- ‘THERMAL RESISTANCE’ (as measured in temperature per unit of power)

-PṬKŘ- ‘THERMAL CONDUCTIVITY’ (as measured in power per distance times temperature)

-ŇŠP- 'ELECTRO-MAGNETIC PHENOMENON'		
BSC	**Stem 1:** (to be) a state of there being an active electric current	**Stem 2:** (to be) a state/act of measuring (or observing a measurement of) an electric current
CTE	(to be) the state of there being an active electric current	
CSV	(to be) a physical act of generating/maintaining/manipulating an electric current; to generate/maintain/manipulate an electric current	**Stem 3:** (to be) a particular system of measuring an electric current; to utilize a particular system of electric current measurement
OBJ	(to be) the actual electric current itself	

The following roots have the same Specification pattern as the root -**ŇŠP**- above:

-**ŇŠPW**- 'ELECTRICAL CHARGE' as measured by electric current times unit of time

-**ŇŠPY**- 'ELECTRICAL POTENTIAL DIFFERENCE AND ELECTROMOTIVE FORCE' as measured in mass times distance^2 divided by unit of current multiplied by time^3

-**ŇŠPL**- 'ELECTRICAL CAPACITANCE' as measured by time^4 times current^2 per mass times distance^2

-**ŇŠPR**- 'ELECTRICAL INDUCTANCE' as measured by mass times distance^2 divided by time^2 times current^2

-**ŇŠPŘ**- 'ELECTRICAL RESISTANCE OF CIRCUITS' as measured by mass times distance^2 divided by the quantity of time^3 times current^2

-**ŇŠPF**- 'ELECTRICAL CONDUCTANCE' as measured by time^3 times current^2 divided by the quantity of mass times distance^2

-**ŇŠPŢ**- 'ELECTRIC FIELD' as measured by force per unit charge or electrical potential difference per distance

-**ŇŠPÇ**- 'ELECTRIC DISPLACEMENT FIELD, POLARIZATION VECTOR' as measured by charge per distance squared

-**MŠPW**- 'ELECTRIC CHARGE DENSITY' as measured by charge per distance cubed

-**MŠPY**- 'ELECTRIC CURRENT DENSITY' as measured by current per distance squared

-**MŠPL**- 'ELECTRICAL RESISTIVITY' as measured by electrical resistance times distance

-**MŠPR**- 'ELECTRICAL CONDUCTIVITY' as measured by electrical conductance per distance

-**MŠPŘ**- 'ELECTROMAGNETIC EMITIVITY' as measured by electrical capacitance per distance

-**MŠPF**- 'ELECTROMAGNETIC PERMEABILITY' as measured by electrical inductance per distance

-**MŠPŢ**- 'ELECTRIC ELASTANCE' as measured by the reciprocal of electrical capacitance

-**LŠP**- 'MAGNETIC FLUX' as measured by energy per unit current = mass times distance^2 divided by time^2 times current

-**LŠPW**- 'MAGNETOMOTIVE FORCE' as measured by a unit of current flowing in a single-turn loop in a vacuum

-**LŠPY**- 'MAGNETIC FLUX DENSITY or B-MAGNETIC FIELD STRENGTH' as measured by force per electric current per distance

-**LŠPL**- ‘H-MAGNETIC FIELD STRENGTH’ as measured by current per distance

-**LŠPR**- ‘EXPOSURE TO IONIZING RADIATION’ as measured by charge per mass

-**LŠPŘ**- ‘RELATIVE DIFFERENCE’ (i.e., ratio between two values of a physical quantity, e.g. power, intensity, current, voltage, loss or gain of an electronic signal, etc.)

-**LŠPF**- ‘RECIPROCAL INDUCTANCE, RELUCTANCE’ as measured by the reciprocal of mass times distance^2 divided by time^2 times current^2

The above roots utilize the following affixes:

-gc	**EMA Electro-Magnetic Measurement A**
1	(in) amperes
2	(in) franklins
3	(in) volts
4	(in) farads
5	(in) coulombs
6	(in) unit poles
7	(in) webers
8	(in) maxwells (or lines)
9	(in) henrys

-gč	**EMB Electro-Magnetic Measurement B**
1	(in) ohms
2	(in) ampere-turns
3	(in) siemens (or mhos)
4	(in) gilberts
5	(in) teslas
6	(in) nepers
7	(in) darafs (i.e., reciprocal of farad)
8	(in) oersteds (i.e., amperes per meter)
9	(in) ohm meters

-gẓ	**EMC Electro-Magnetic Measurement C**
1	(in) volts per meter
2	(in) newtons per coulomb
3	(in) coulombs per sq. meter
4	(in) coulombs per cu. meter
5	(in) amperes per sq. meter
6	(in) siemens per meter
7	(in) farads per meter
8	(in) henrys per meter
9	(in) coulombs per kilogram

Additional roots with the above Specification pattern:

-**RŠP**- ‘LUMINOUS INTENSITY’ i.e., power emitted by a light source in a particular direction per unit solid angle as measured in candelas

-**RŠPW**- ‘LUMINANCE’ as measured by luminous intensity per distance squared

-**RŠPY**- ‘LUMINOUS FLUX’ i.e. “amount” of visible light emitted by a source, as measured by luminous intensity per solid angle

-**RŠPL**- ‘ILLUMINANCE, LUMINOUS EXITANCE OR EMITTANCE’ as measured by luminous intensity per solid angle per distance squared

-**RŠPR**- ‘PHOTON FLUX, AIRGLOW’ as measured in photons per distance squared per unit of time per solid angle

-**RŠPŘ**- ‘LUMINOUS ENERGY’ i.e., the perceived energy of light, as measured by luminous intensity times unit of time per solid angle (e.g., talbots)

-**RŠPF**- ‘LUMINOUS ENERGY DENSITY’ as measured by luminous intensity times unit of time per solid angle per cubic distance

-**RŠPŢ**- ‘LUMINOUS EXPOSURE’ as measured by luminous intensity times unit of time per distance squared (e.g., lux second)

-**RŠPÇ**- ‘LUMINOUS EFFICACY’ as measured by luminous intensity per solid angle per unit power (e.g., lumen per watt)

Associated affixes for the above group of roots:

-fẓ	LOA Luminance & Optics A
1	(in) candelas
2	(in) candle-powers
3	(in) lumens (= candela steradians)
4	(in) foot-candles
5	(in) lux
6	(in) nits (= candelas per square meter)
7	(in) talbots
8	(in) rayleighs
9	(in) lux seconds

-fj	LOB Luminance & Optics B
1	(in) lumens per square meter
2	(in) lumen seconds per cubic meter
3	(in) lumens per watt
4	(in) lamberts
5	(in) foot-lamberts
6	(in) stilbs
7	(in) diopters
8	(in) skots
9	(in) brils

Additional roots with the above Specification pattern:

-**GZK**- 'DENSITY' as measured by mass per volume

-**GZKW**- 'CATALYTIC ACTIVITY' as measured by amount of substance per unit time

-**GZKY**- 'SUBSTANCE CONCENTRATION' as measured by amount of substance per cubic distance

-**GZKL**- 'ENERGY PER AMOUNT OF SUBSTANCE'

-**GZKR**- 'VOLUME OCCUPIED BY AN AMOUNT OF A SUBSTANCE AT A GIVEN TEMPERATURE AND PRESSURE' as measured in cubic distance per amount of substance

-**GZKŘ**- 'MOLAR HEAT CAPACITY, MOLAR ENTROPY' i.e., ratio of the heat added to (or subtracted from) an object to the resulting temperature change, as measured in energy per unit temperature times amount of substance

-**GZKF**- 'MOLAR CONDUCTIVITY' i.e., efficiency of electrical conductivity of a substance, as measured by electrical conductance times square distance per amount of substance

-**GZKŢ**- 'CHEMICAL CONCENTRATION' as measured by amount of substance per unit mass

Associated affix with the above group of roots:

-fč	DMU Density-Releated Measurement Units
1	(in) moles
2	(in) International Units (IU)
3	(in) katals (= moles per second)
4	(in) moles per cubic meter
5	(in) joules per mole
6	(in) molar volumes (= cubic meter per mole)
7	(in) molals (= mole per kilogram)
8	(in) joules per kelvin mole
9	(in) siemens times sq. meter per mole

-X- 'SIZE / MEASURE / MEASURED DURATION' (use SUF/EXN/EXD, etc. affixes to specify degree)		
BSC	**STEM 1:** (to be/manifest a) degree of spatio-temporal size (= the amount/volume of space or time taken up by an entity)	**STEM 2:** (to be/manifest a) degree of spatial size, i.e., volume of space (whether uni-dimensional, 2-D, or 3-D) **STEM 3:** (to be/manifest a) degree of temporal "size" (= an "amount" of time)
CTE	(to be) something with a size [focus on the entity itself]	
CSV	(to be) a size (= the volume of space or time taken up)	
OBJ	(to be) the particular size of an entity; to measure the size of an entity	

Morphological derivations: enlarge, swell, expand (in volume), grow (in size); shrink, make smaller

-RF- PORTION / AMOUNT / RATION / ALLOTMENT Associated Affix: **PTW**		
BSC	**Stem 1:** (to be) an amount/quantity of a non-countable substance/entity [both the entity/substance and its amount]; to determine/fix/distribute an amount/quanity of something	**Stem 2:** (to be) a share, a portion (to be) distributed **Stem 3:** (to be) an allotment/allocation, set distribution
CTE	(to be) the entity/substance contained in or comprising an amount/quantity	
CSV	(to be) an (ostensibly or theoretically measurable) amount/quantity; to determine/fix/distribute/disperse a particular amount/quantity	
OBJ	(to be) the object/entity made of/consisting of an amount/quantity of something	

Derivations: to share, distribute, allocate, allot

The Specification pattern of the above root is also used for the stems of the following root:

-RFW- ‘DOSE / PRESCRIBED AMOUNT / RATION / APPORTIONMENT / QUOTA’
Stem 1: (to be a) measure of, dose of, a formally pre-determined amount of a substance/entity **Stem 2:** (to be) an official/authorized share/ration, doled-out amount **Stem 3:** (to be) an official/authorized allotment/allocation/quota

6.3 PHYSICS AND COSMOLOGICAL/ASTRONOMICAL/GEOPHYSICAL PHENOMENA

-RSW- ‘PLANETARY BODY’		
BSC	**Stem 1:** (to be) a planet/planetary body; to travel one’s orbit in space as a planet	**Stem 2:** (to be) a moon of planetary body [i.e., sufficiently large to be spherical in shape]; to travel one’s orbit in space as a moon/satellite of a planet **Stem 3:** (to be) a large (spherical) asteroid or micro-planet
CTE	(to be) a world [i.e., the planetary home of a holistic eco-system/biosphere, etc.]	
CSV	(to be) a physical planetary body	
OBJ	(to be) the orbit(al path) of a planetary body; to orbit	

-SW- ‘EARTH / MOON / NAMED MICRO-PLANET’		
BSC	**STEM 1:** (to be) the Earth/Terra (as both a planetary body and a “world” encompassing a particular biosphere/ecosystem and civilization)	**STEM 2:** (to be) the Moon/Luna **STEM 3:** (to be) one of the named asteroids or micro-planets of the solar system (e.g., Pluto, Ceres, Juno, Sedna, etc.)
CTE	(to be) “the world” [i.e., the Terran-based biosphere, ecosystem, environment and civilizaton which Terran life inhabits]	
CSV	(to be) the physical planetary body itself that constitutes the planet Earth/Terra	
OBJ	(to be) the orbit(al path) of the planet Earth/Terra	

-LXW- 'NON-PLANETARY ASTRAL BODY'		
BSC	**Stem 1:** (to be) a stellar body / star	**Stem 2:** (to be) a comet **Stem 3:** (to be) a small [non-spherical] asteroid)
CTE	(to be) a state of a star shining	
CSV	(to be) the physical body/mass of a star	
OBJ	(to be) the location in space of a star (relative to other astral bodies)	

-LX- 'THE SUN (SOL) / NAMED COMET / NAMED PLANETESIMAL or ASTEROID'		
BSC	**STEM 1:** (to be) the Sun / Sol	**STEM 2:** (to be) a named comet **STEM 3:** (to be) a named small asteroid/planetesimal (e.g., Eros)
CTE	(to be) the functional (shining) state of the Sun/Sol; for there to be sunshine, (for the Sun) to shine	
CSV	(to be) the physical body/mass of the Sun/Sol	
OBJ	(to be) the location in space of the Sun/Sol relative to other astral bodies	

-MTŘ- 'NEBULA'		
BSC	**Stem 1:** (to be) a gas-based emission-type nebula or volume of nebulosity in interstellar space (i.e. cloud in space of of high-temperature and/or ionized gas, e.g., an H II region or Herbig-Haro object)	**Stem 2:** (to be) a reflection nebula or dark nebulae (i.e., cloud of dust in space, whether reflecting or blocking light from another source) **Stem 3:** (to be) a planetary nebula (i.e., gaseous cloud thrown out by a star at the end of its life) CPT = supernova remnant nebula
CTE	(to be) a state of a gaseousness/nebulosity in a gas-based nebula	
CSV	(to be) the physical body/mass constituting a gas-based nebula	
OBJ	(to be) the location in space of a (relative to other celestial bodies) of a gas-based nebula	

-MTW- 'CELESTIAL SPACE / CELESTIAL MEDIUM'		
BSC	**Stem 1:** (to be) a volume of celestial (i.e., interplanetary, interstellar/intergalactic) space; to be (situated) in space	**Stem 2:** (to be) a naturally occurring physical material filling or contained within a volume of celestial space (e.g., cosmic dust, plasma or ionized clouds of hydrogen or helium, etc.) **Stem 3:** (to be) a magnetic or energy-based field occurring in or contained within a volume of celestial space
CTE	(to be) the state/quality of three-dimensional "spaciousness" or "room" within a volume of celestial space	
CSV	(to be) the physical body/structure/medium of celestial space; to occur or be situated in celestial space	
OBJ	(to be) a location in space (relative to celestial bodies within that space)	

-LXṬ- 'ATYPICAL COMPACT STAR'		
BSC	**Stem 1:** (to be) a black hole	**Stem 2:** (to be) a neutron star (including pulsar and magnetar) **Stem 3:** (to be) a white dwarf star
CTE	(to be) a state/quality associated with the unusual physics of a black hole	
CSV	(to be) the physical body/mass of a black hole	
OBJ	(to be) the location in space of a black hole (relative to other celestial bodies)	

-KKW- 'ATOM'		
BSC	**Stem 1:** (to be) an atom	**Stem 2:** (to be) the electron cloud of an atom (i.e., set of electron shells with "orbiting" electrons) of an atom **Stem 3:** (to be) an orbital state (i.e., state and behavior of an electron based on its orbital probability distribution)
CTE	(to be) the state of an atom in terms of its interaction (or lack thereof) with other atoms; for an atom to interact (via atomic bonding) with other atoms	
CSV	(to be) the tangible/physical "embodiment"/manifestation of an atom	
OBJ	(to be) a component/attribute/aspect of an atom (i.e., a particle, a charge, a force, etc.)	

-KKY- 'ATOMIC NUCLEUS'		
BSC	**Stem 1:** (to be) the nucleus of an atom	**Stem 2:** (to be) a nuclide (i.e., specific configuration of Z-value, N-value, and energy state for a particular atomic nucleus) **Stem 3:** (to be) a state/process of nucleosynthesis resulting in an atomic nucleus; CTE = state which gives rise to nucleo-synthesis; CSV = the physical process itself of nucleosynthesis; OBJ = the atomic nucleus resulting from a process of nucleosynthesis
CTE	(to be) the state of an atomic nucleus in terms of its interaction (or lack thereof) with other atoms; for an atom to interact (via atomic bonding) with other atoms	
CSV	(to be) the tangible/physical "embodiment"/manifestation of an atomic nucleus	
OBJ	(to be) a component/attribute/aspect of an atomic nucleus (i.e., a particle, a charge, a force, etc.)	

-LṬK- 'SUB-ATOMIC PARTICLE'		
BSC	**Stem 1:** (to be) a sub-atomic particle	**Stem 2:** (to be) an elementary particle (i.e., fermion or boson) **Stem 3:** (to be) a composite particle (i.e., a hadron, whether baryon or meson)
CTE	(to be) the sub-atomic state or "force" (i.e., interaction) mediated or maintained by a sub-atomic particle	
CSV	(to be) the tangible/physical "embodiment"/manifestation/"stuff" of a sub-atomic particle	
OBJ	(to be) an attribute of sub-atomic particle (e.g., mass, charge, spin, angular momentum, etc.)	

The following roots follow the Specification pattern of the above root:

-**LṬKW**- 'PHOTON / GLUON / GRAVITON' (Electromagnetism, Strong Nuclear Force, Gravity) 1. photon 2. gluon 3. graviton

-**LṬKY**- 'ELECTRON / PROTON / NEUTRON' 1. electron 2. proton 3. neutron

-**LṬKL**- 'NEUTRINO' 1. electron neutrino 2. muon neutrino 3. tau neutrino

-**LṬKR**- 'FERMION' 1. fermion 2. quark 3. lepton

-**LṬKŘ**- 'BOSON' 1. boson 2. "string" (from string theory) [CTE Specification = vibration of "string"] 3. acceleron (i.e., hypothetical particle associated with dark energy theory)

-**LṬKF**- 'ANTI-FERMION' 1. anti-fermion 2. anti-quark 3. anti-lepton

-**LṬKṬ**- 'HIGGS BOSON / MUON / TAUON' 1. Higgs boson 2. muon 3. tauon (i.e., tau lepton)

-**LṬKÇ**- 'WEAK GAUGE BOSON (Weak Nuclear Force) 1. W+ boson 2. W- boson 3. Z-zero boson

-**LṬKM**- 'BARYON' 1. nucleon 2. hyperon 3. charmed particle

-**LṬKN**- 'MESON' 1. vector meson 2. pseudoscalar meson 3. light-unflavored meson

'OTHER SUB-ATOMIC PARTICLE' I 1. pion 2. kaon 3. theta meson

'OTHER SUB-ATOMIC PARTICLE' II 1. rho meson 2. eta meson 3. D-meson

'OTHER SUB-ATOMIC PARTICLE' III 1. B-meson 2. omega meson 3. phi meson

'OTHER SUB-ATOMIC PARTICLE' IV 1. psion 2. upsilon meson 3. sub-atomic particle other than those listed above

Affixes available for some of the above stems:

-xj	**ASP Attributes of Sub-Atomic Particles**
1	having opposite charge, antimatter version of X
2	super-symmetrical partner of X, "sparticle" counterpart of X , e.g., photino, gluino, gravitino, slepton, squark, sneutrino, etc.
3	spin value (intrinsic angular momentum) of particle X
4	orbital angular momentum of particle X
5	mass of particle X
6	isospin of particle X
7	parity of particle X
8	G-parity of particle X
9	C-parity of particle X

-bj	CFQ "Colors" and "Flavors" of Quarks
1	up
2	charm
3	top
4	red
5	green
6	blue
7	bottom
8	strange
9	down

-LṬW- 'WAVE / WAVE-FRONT / STANDING WAVE'		
BSC	**Stem 1:** (to be) a mechanical wave; to move/propogate/act as a mechanical wave (e.g., sound, water, seismic, surface, string vibration, etc.) [i.e., propogated oscillations through a medium]	**Stem 2:** (to be) an electromagnetic or other non-mechanical wave; to move/propogate/act as an electro-magnetic wave (e.g., visible light, radio, ultraviolet, infrared, x-ray, gamma ray, gravitational wave, reaction-diffusion wave, quantum probability wave, etc.) **Stem 3:** (to be) a standing wave (e.g., a plucked string of a violin vibrating); to vibrate as a standing wave
CTE	(to be) the state of oscillation/vibration which makes up a wave	
CSV	(to be) an act/process of generating a wave; to generate a wave	
OBJ	(to be) a wave-front, the leading edge of a wave phenomenon	

-VPR- 'ELECTRICITY / MAGNETISM / IONIZATION'		
BSC	**Stem 1:** (to be) an act/instance/occurrence of electricity flowing	**Stem 2:** (to be) an act/instance/occurrence of magnetism, magnetic alignment, magnetic force **Stem 3:** (to be) a act/instance/occurrence of ionization; to ionize
CTE	(to be) the state of electrical current flow	
CSV	(to be) an act/instance/occurrence of an electrical current or discharge; for electricity to flow or discharge	
OBJ	(to be) an object/entity acted upon by electricity	

-SH- 'AIR'		
BSC	**Stem 1:** (to be) the state/act/process of there being air in a particular spatio-temporal location (i.e., the standard combination of nitrogen/oxygen/argon/CO_2, etc. that constitutes the breathable atmosphere of Earth	**Stem 2:** (to be) the state/act/process of there being "fresh" air (i.e., air let in from an external source/outside to replenish the recycled or "stale" air in a location) **Stem 3:** (to be) the state/act/process of there being "stale" air (i.e., air with a lowered oxygen content and/or higher CO2 concentration and/or pollutants due to breathing in an enclosed space, air pollution, etc.)
CTE	(to be) air as the primary/relevant component of Earth's (or a planet's) atmosphere	
CSV	(to be) an act of obtaining/providing/disseminating air (into a volume of space); to get air, give air, let air in	
OBJ	(to be) the air one breathes	

-CHW- 'OUTDOORS / OUTSIDE / INDOORS'		
BSC	**Stem 1:** (to be) the state/act of being/occurring/manifesting outdoors, in the open air, outside; to be/occur/manifest outdoors, be/take place in the open air	**Stem 2:** (to be) the state/act/process of being/occurring/manifesting indoors/inside, within the enclosed space(s) of an interior environment separate from the outdoors **Stem 3:** (to be) the state/act/process of being/occurring/manifesting in a protected/localized interior/enclosed environment while outdoors (e.g., in a spacesuit on a planetary surface, in a transparent tent outdoors, in a screened-off sun-room of a house, on the screen-enclosed balcony/ledge of a building)
CTE	(to be) a state of being in an outdoor, open-air setting	
CSV	(to be) a physical act/process of making/letting something happen outdoors	
OBJ	(to be) the outdoors, the open air, outside (as a location/setting/environment for something to be or take place in)	

-KTL- 'AREA / AREAL FOOTPRINT / PLOT OF GROUND / PLOT OF LAND'		
BSC	**Stem 1:** (to be) the quantitative area of a bounded quasi-planar surface, i.e., the amount of square surface space contained within a bounded area (including area expanded by non-level variations in surface gradients); to measure the area of a bounded quasi-planar surface	**Stem 2:** (to be) the quantitative surface-footprint or areal footprint taken up by a quasi-planar surface regardless of non-level variations in surface gradients **Stem 3:** (to be) a formal/bounded plot of ground/land based on natural or authorized delineations
CTE	(to be) the physical ground/surface of a bounded area	
CSV	(to be) the act of measuring/determining the quantitative area of a bounded quasi-planar surface	
OBJ	(to be) the actual measured amount of area (i.e., how much acreage/square meters, etc.)	

-ÇB- 'TERRESTRIAL TOPOGRAPHY / GEOGRAPHIC FEATURE / ARABILITY'		
BSC	**Stem 1:** (to be) a terrestial/geographic land feature	**Stem 2:** (to be) an area of arable land **Stem 3:** (to be) an area of non-arable land
CTE	(to be) a particular geographic/terrestrial feature itself	
CSV	(to be) the particular aspects/characteristics of a particular type of terrestrial land feature (e.g., the aridness/starkness of a desert, the packed ice/snow of a glacier, the vastness/flatness of a the plains/pampas, etc.)	
OBJ	(to be) the geographic/ecological impact of a geographic feature on its surrounding environment	

-JL- 'MOUNTAINOUS TOPOGRAPHY' [use appropriate affixes to indicate degree of relative elevation]		
BSC	**Stem 1:** (to be) a terrestrial land feature of raised elevation in relation to sea level or surrounding flatlands; a hill or mountain	**Stem 2:** (to be) a peak, a single elevated terrestrial upwelling higher than the surrounding the terrain **Stem 3:** (to be) a plateau/butte/mesa (i.e., a high/elevated flatland)
CTE	(to be) the state of being a hill/mountain	
CSV	(to be) a state of dealing with mountainous/elevated terrain; to deal with (e.g., traveling through, living in, etc.) mountainous/elevated terrain	
OBJ	(to be) the geographic/ecological impact of mountain on its surrounding environment	

-ZH- 'DESERT / STEPPE / TUNDRA'		
BSC	**Stem 1:** (to be) a desert	**Stem 2:** (to be) an area of steppe, steppe-like terrain **Stem 3:** (to be) an area of tundra, tundra-like terrain
CTE	(to be) the state of being a desert	
CSV	(to be) a state of dealing with desert terrain; to deal with (e.g., traveling through, living in, etc.) desert terrain	
OBJ	(to be) the geographic/ecological impact of a desert on its surrounding environment	

-VŇ- 'VALLEY / CANYON / GORGE'		
BSC	**Stem 1:** (to be) a valley	**Stem 2:** (to be) a canyon **Stem 3:** (to be) a gully
CTE	(to be) the state of being a valley	
CSV	(to be) a state of dealing with a valley as a type of terrain; to deal with (e.g., traveling through, living in, etc.) a valley	
OBJ	(to be) the geographic/ecological impact of a valley on its surrounding environment	

Derivation: gorge

-CTH- 'CONTINENT / TECTONIC PLATE / LAND MASS'		
BSC	**Stem 1:** (to be) a continent [i.e., raised land mass taking up a significant percentage of a planetary surface, mostly or entirely surrounded by ocean]	**Stem 2:** (to be) a tectonic plate **Stem 3:** (to be) a land mass
CTE	(to be) the state of being a continent	
CSV	(to be) a state of utilizing/living on a continent; to deal with living on a coninent	
OBJ	(to be) the geographic/ecological impact of a continent on its surrounding environment	

-CKH- 'ISLAND / PENINSULA / ISTHMUS'		
BSC	**Stem 1:** (to be) an island	**Stem 2:** (to be) a peninsula **Stem 3:** (to be) an isthmus
CTE	(to be) the state of being an island	
CSV	(to be) a state of utilizing/living on an island; to deal with living on an island	
OBJ	(to be) the geographic/ecological impact of an island on its surrounding environment	

Derivations: islet, cape, land-bridge

-ŽPĻ- 'GEOTHERMAL FEATURE'		
BSC	**Stem 1:** (to be) a volcano	**Stem 2:** (to be) a hot spring **Stem 3:** (to be) a geothermal vent
CTE	(to be) the state of being a volcano	
CSV	(to be) a state of with living on or dealing with a volcano	
OBJ	(to be) the geographic/ecological impact of a volcano on its surrounding environment	

-CGL- 'ICICLE / RUSTICLE / BRINICLE '		
BSC	**Stem 1:** (to be) a process of forming/becoming an icicle; to form an icicle CPT = a formed/longstanding icicle	**Stem 2:** (to be) a process of forming/becoming a rusticle; to form a rusticle **Stem 3:** (to be) a process of forming/becoming a brinicle; to form a brinicle
CTE	(to be) the state of being an icicle	
CSV	(to be) the process of icicle formation; to physically form an icicle by drop-by-drop accumulation of water in the process of freezing	
OBJ	(to be) the geographic/ecological impact of an icicle on its surrounding environment	

The Specification pattern of the above root applies to the following roots/stems as well:

-CGR- 1. stalactite 2. stalagmite 3. bottlebrush
-CGW- 1. lavacicle 2. karst 3. tufa mound

-ṬR- 'WATER / SOURCE OF WATER		
BSC	**Stem 1:** (to be) an amount of water; to water, pour/draw/procure some water	**Stem 2:** (to be) a natural source of fresh water (e.g., a natural well, river, spring, etc.) **Stem 3:** (to be) a a human-made source of fresh water, i.e., a cistern, a reservoir, etc.
CTE	(to be) some water (as a substance)	
CSV	(to be) a state/act of drawing/pouring/filling/serving an amount of water; to draw/pour/fill/serve some water	
OBJ	(to be) the container, vessel, conveying implement which holds an amount of water	

-**RṬR**- **'WATER IN MOTION OR USE: FLOW / IRRIGATION / CASCADE / WATERFALL'**		
BSC	**Stem 1:** (to be) a state/act of water running or flowing; to run/flow [in reference to water or, by analogy, any liquid/fluid with water-like consistency]	**Stem 2:** (to be) state/act of utilizing water, irrigation, watering; to water / irrigate **Stem 3:** a waterfall, a cascade
CTE	(to be) a state of flowing; to be flowing	
CSV	(to be) an act of letting/making flow; to make/let flow	
OBJ	(to be) a flow of water [i.e., a particular amount of flowing water]	

-BW- 'BODY OF SALT WATER'		
BSC	**Stem 1:** (to be) an ocean/sea	**Stem 2:** (to be) a lagoon **Stem 3:** (to be) a saltwater pond/lake
CTE	(to be) the state of being an ocean/sea	
CSV	(to be) a state of utilizing/living by the ocean/sea; to deal with, live by, utilize the ocean/sea	
OBJ	(to be) a particular (i.e., named) ocean/sea	

-JY- 'WATER CHANNEL BETWEEN LAND MASSES / BAY / INLET'		
BSC	**Stem 1:** (to be) a channel-like body of water allowing passage between two land masses	**Stem 2:** (to be) a bay [i.e., body of water partially surrounded by land] **Stem 3:** (to be) a narrow river-like inlet (on a body of water)
CTE	(to be) the state of being a channel-like body of water	
CSV	(to be) a state of utilizing/living by channel-like body of water; to deal with, live by, utilize a channel-like body of water	
OBJ	(to be) a particular (i.e., named) channel-like body of water	

Derivations: strait, gulf, bight, fjord

-LZ- 'BODY OF FRESH WATER' [use appropriate affixes to specify size/volume]		
BSC	**Stem 1:** (to be) a pond/lake	**Stem 2:** (to be) a stream/river **Stem 3:** (to be) a freshwater inland sea
CTE	(to be) the state of being a pond/lake	
CSV	(to be) a state of utilizing/living by a pond/lake; to deal with, live by, utilize a pond/lake	
OBJ	(to be) a particular (i.e., named) pond/lake	

-LZW- 'ESTUARY / RIVER DELTA / FRESHWATER-SALTWATER TRANSITION ZONE'		
BSC	**Stem 1:** (to be) an estuary	**Stem 2:** (to be) a river delta **Stem 3:** (to be) a freshwater-saltwater transition zone (e.g., the mouth of the Amazon or Mississippi Rivers)
CTE	(to be) the state of being an estuary	
CSV	(to be) a state of utilizing/living by an estuary; to deal with, live by, utilize an estuary	
OBJ	(to be) a particular (i.e., named) estuary	

-LZR- 'SWAMP / MARSH / BOG / FEN / BAYOU / WETLANDS'		
BSC	**Stem 1:** (to be) a swamp/marsh	**Stem 2:** (to be) a bog/fen **Stem 3:** (to be) a bayou/wetland(s)
CTE	(to be) the state of being a swamp/marsh	
CSV	(to be) a state of utilizing/living by a swamp/marsh; to deal with, live by, utilize a swamp/marsh	
OBJ	(to be) a particular (i.e., named) swamp	

-ÇTY- ‘NORTH / SOUTH / LONGITUDE’		
BSC	**Stem 1:** (to be) a state of north-south directional orientation	**Stem 2:** (to be) a state/act of being northern or in/of/toward the northern part/area of something in terms of (relative) location/place/origin, etc. **Stem 3:** (to be) a state/act of being southern or in/of the southern part/area of something in terms of (relative) location/place/origin, etc.
CTE	(to be) a north-south (i.e., longitudinal) line/path/orientation	
CSV	(to be) a state/act of lying upon, traveling upon, or being directed along a north-south, i.e., longitudinal axis	
OBJ	(to be) what lies upon or is moving or being directed upon a longitudinal line/axis/path, etc.	

-DNY- ‘EAST / WEST / LATITUDE’		
BSC	**Stem 1:** (to be) a state of east-west directional orientation	**Stem 2:** (to be) a state/act of being eastern or in/of/toward the eastern part/area of something in terms of (relative) location/place/origin, etc. **Stem 3:** (to be) a state/act of being western or in/of the western part/area of something in terms of (relative) location/place/origin, etc.
CTE	(to be) an east-west (i.e., latitudinal) line/path/orientation	
CSV	(to be) a state/act of lying upon, traveling upon, or being directed along a east-west, i.e., latitudinal axis	
OBJ	(to be) what lies upon or is moving or being directed upon a latitudinal line/axis/path, etc.	

-NTŇ- ‘LATITUDINAL BASES [LATITUDINAL POLE / EQUATOR]’		
BSC	**Stem 1:** (to be) a state/act of being situated or occurring on/at the equatorial line of latitude; be/occur on/at/along the equator	**Stem 2:** (to be) a state/act of being situated or occurring on/at the [geographic, not magnetic] north pole **Stem 3:** (to be) a state/act of being situated or occurring on/at the [geographic, not magnetic] south pole
CTE	(to be) the equator	
CSV	(to be) an act of situating something or being situated at/on/along the equator; to happen or make happen at/on/along the equator	
OBJ	(to be) what is situated or takes place on/at the equator	

-VS- ‘SEASON’		
BSC	**Stem 1:** (to be) a particular season (i.e., spring, summer, autumn, winter)	**Stem 2:** (to be) the natural condition or state of growth/dormancy of Nature’s flora, as associated with a particular season **Stem 3:** (to be) the solstice or equinox associated with a particular season; to be the first day of a season as determined by it being a solstice or equinox
CTE	(to be) the state of it being a certain season; be seasonal	
CSV	(to be) a phenomenon occurring during a particular season	
OBJ	(to be) a seasonal activity/event/state, i.e., something that happens during a particular season; be something seasonal	

-ZY- ‘SPRING (SEASON) / SPRING-RELATED NATURAL PHENOMENON’		
BSC	**Stem 1:** (to be) Spring, springtime	**Stem 2:** (to be) the blooming/flowering/growing phenomenon of Nature’s flora occurring/due to it being springtime **Stem 3:** (to be) the Spring equinox
CTE	(to be) the state of it being springtime	
CSV	(to be) a phenomenon/event/state/act occurring during (or because it is) springtime; to occur during (or because it is) springtime	
OBJ	(to be) an activity one does/performs during/because it is springtime	

-MŘ- ‘SUMMER (SEASON) / SUMMER-RELATED NATURAL PHENOMENON’		
BSC	**Stem 1:** (to be) Summer, summertime	**Stem 2:** (to be) the peak state/condition of growth/vibrancy/life of Nature’s flora occurring/due to it being summertime **Stem 3:** (to be) the Summer solstice
CTE	(to be) the state of it being summertime	
CSV	(to be) a phenomenon/event/state/act occurring during (or because it is) summertime; to occur during (or because it is) summertime	
OBJ	(to be) an activity one does/performs during/because it is summertime	

-ŇŠ- ‘AUTUMN (SEASON) / AUTUMN-RELATED NATURAL PHENOMENON’		
BSC	**Stem 1:** (to be) Autumn, fall	**Stem 2:** (to be) the withering/dying phenomenon of Nature’s flora occurring/due to it being autumn **Stem 3:** (to be) the Autumn equinox
CTE	(to be) the state of it being autumn	
CSV	(to be) a phenomenon/event/state/act occurring during (or because it is) autumn; to occur during (or because it is) autumn	
OBJ	(to be) an activity one does/performs during/because it is autumn	

-ÇČ- ‘WINTER (SEASON) / WINTER-RELATED NATURAL PHENOMENON’		
BSC	**Stem 1:** (to be) Winter, wintertime	**Stem 2:** (to be) the state/condition of dormancy/lifelessness of Nature’s flora occurring/due to it being wintertime **Stem 3:** (to be) the Winter solstice
CTE	(to be) the state of it being wintertime	
CSV	(to be) a phenomenon/event/state/act occurring during (or because it is) wintertime; to occur during (or because it is) wintertime	
OBJ	(to be) an activity one does/performs during/because it is wintertime	

-VSL- ‘SEASONALLY-DETERMINED SOLAR EXPOSURE OR TEMPERATURE’		
BSC	**Stem 1:** (to be) a state of the ambient outdoor temperature being in a certain range due to the season of the year; to be in a seasonal temperature range	**Stem 2:** (to be) the duration of daylight during a 24-hour period in a location due to the time (i.e., season) of the year **Stem 3:** (to be) the angle of the sun (i.e., height of the arc of the sun’s passage) due to the time (i.e., season) of the year
CTE	(to be) a state of it being a certain seasonally-determined temperature	
CSV	(to be) an act of determining the ambient temperature range; to see/feel how hot/cold it is outside (based on the expected temperature range of the season)	
OBJ	(to be) the ambient temperature (as expected for the season)	

-FŠL- ‘WEATHER’		
BSC	**Stem 1:** (to be) a weather condition; to be the weather (conditions)	**Stem 2:** (to be) a state of good weather conditions **Stem 3:** (to be) a state of adverse weather condition
CTE	(to be) the actual extant weather condition at a particular time	
CSV	(to be) the physical manifestation(s) of a particular weather condition (e.g., the rain itself, the fog itself, the sunniness itself, the heat itself, the wind itself, etc.)	
OBJ	(to be) a party effected/impacted by the weather conditions	

The following root has the same Specification pattern as the root immediately above:

-FŠR- ‘CLIME / CLIMATE’
Stem 1: (to be) the climate, clime **Stem 2:** (to be) a cold climate **Stem 3:** (to be) a warm climate

-FTH- ‘SKY / ATMOSPHERE / ATMOSPHERIC PHENOMENON / ATMOSPHERIC PRESSURE / CLOUD’		
BSC	**Stem 1:** (to be) a state of there being a gaseous atmosphere enveloping a planetary body; to be a planetary atmosphere	**Stem 2:** (to be) the state/act/process of there being a particular amount of air pressure in the atmosphere; to be a certain atmospheric pressure [i.e., at any given altitude) **Stem 3:** (to be) the state/act/process of there being a cloud (in the sky); to be a cloud (in the sky, for a cloud (in the sky) to be present
CTE	(to be) the collection of gaseous elements which comprise an atmosphere	
CSV	(to be) a process/phenomenon of the atmosphere; to manifest an atmospheric phenomenon	
OBJ	(to be) the enveloping gaseous, visible (via reflected sunlight) medium of the atmosphere itself, i.e., the sky; to be the sky	

NOTE: for ‘fog; to be foggy’, concatenate Stem 2 of -**KTH**- (‘ground/surface of Earth’) in LOCATIVE format into Stem 3 of the above root (‘cloud’)

The following root has the same Specification pattern as the root immediately above:

-TTH- 'CLOUD TYPE'
Stem 1: cumuliform cloud **Stem 2:** stratiform cloud **Stem 3:** cirriform cloud

Use stem concatenation distinguish combinations such as stratocumulus, cirrocumulus, and cirrostratus, or clouds classed by shape, e.g., nimbus

-MṬR- 'WATER-BASED WEATHER PHENOMENON'		
BSC	**Stem 1:** (to be) an amount of rainfall; to rain [CPT version = fallen rain, fallen precipitation]	**Stem 2:** (to be) an amount of snow; to snow CPT = fallen snow] **Stem 3:** (to be) an amount of hail; to hail [CPT = ice on the ground]
CTE	(to be) a state of rainfall, for rain to be falling	
CSV	(to be) a state/act of rain falling	
OBJ	(to be) an amount of rainwater (e.g., in a puddle, in a container, in one's hand, etc.)	

NOTE: for 'fog; to be foggy', concatenate Stem 2 of -**KTH**- ('ground/surface of Earth') in LOCATIVE format into Stem 3 of -**FTH**- ('cloud')

-FPH- 'WEATHER FRONT / AIR MASS'		
BSC	**Stem 1:** (to be) a state of there being an air mass/weather front (as differentiated by temperature, water vapor content, and pressure from surrounding air masses)	**Stem 2:** (to be) a state of there being a cold front **Stem 3:** (to be) a state of there being a warm front
CTE	(to be) the air mass itself which comprises a weather front	
CSV	(to be) a state/process of a weather front affecting the weather of a geographic area	
OBJ	(to be) the weather condition caused by a particular weather front	

-FKH- ‘WIND / STORM’		
BSC	**Stem 1:** (to be) a state of there being a momentary air-pressure-based atmospheric phenomenon, to be a gust of wind, for a gust of wind to blow, for an air current to blow	**Stem 2:** (to be) a storm **Stem 3:** (to be) a spirally-oriented storm driven by massive air-pressure variances and Coriolis forces [use appropriate affixes to indicate severity as a tropical depression, hurricane/cyclone/typhoon, tornado, etc.)
CTE	(to be) the collection of air (or other gas) reacting to pressure variance which comprise a gust wind	
CSV	(to be) a process/phenomenon of a gust of wind blowing; for a gust of wind to blow	
OBJ	(to be) the wave-front of a gust of wind (i.e., the concentration of gaseous molecules) which constitutes a (potentially destructive) energetic force which can be felt or which can physically impact objects contacted by the wave front.	

-MHW- ‘HUMIDITY / MUGGINESS / BALMINESS / CRISPNESS / BRISKNESS’		
BSC	**Stem 1:** (to be) a state of humid/muggy weather; to be muggy/humid	**Stem 2:** (to be) a state of balmy, calm, clear, pleasant weather; to be balmy, be calm, be clear, be pleasant weather **Stem 3:** (to be) a state of crisp/brisk weather; to be crisp/brisk weather
CTE	(to be) a state/process of ambient heat combined with high humidity that constitutes mugginess	
CSV	(to be) a state/process of having to deal with, be exposed to muggy weather conditions	
OBJ	(to be) a state of personal discomfort caused by humid/muggy weather	

-VPL- ‘LIGHTNING / ELECTRICAL STORM’		
BSC	**Stem 1:** (to be) a state of there being a lightning strike on something; for lightning to strike something	**Stem 2:** (to be) an electrical storm **Stem 3:** (to be) the ionized atmospheric conditions necessary to produce lightning
CTE	(to be) the electrical charge/voltage/amperage/power of a lightning strike	
CSV	(to be) a process/phenomenon of a bolt of lightening striking; for a bolt of lightning to strike	
OBJ	(to be) a bolt of lightning	

-ŇPĻ- ‘RAINBOW / AURORA / ZODIACAL LIGHT / GEGENSCHEIN’		
BSC	**Stem 1:** (to be) a state of there being a visible rainbow; for there to be a visible rainbow	**Stem 2:** (to be) a state of there being a visible aurora **Stem 3:** (to be) a state of there being sunlight reflected from interplanetary dust particles visible under certain conditions (e.g., zodiacal light (“false dawn”), gegenschein/counterglow)
CTE	(to be) the light passing through aerosolized water vapor that allows a rainbow to be seen	
CSV	(to be) the state of seeing a rainbow; to see a rainbow	
OBJ	(to be) the atmospheric state of aerosolized water vapor conducive to/needed for seeing a rainbow	

6.4 MATHEMATICAL CONCEPTS

6.4.1 The Basic Number Roots

0	1	2	3	4	5	6	7	8	9	10	100	100^2	100^4	100^8
-VR-	**-LL-**	**-KS-**	**-Z-**	**-PŠ-**	**-ST-**	**-CP-**	**-NS-**	**-ČK-**	**-LẔ-**	**-J-**	**-GZ-**	**-PC-**	**-KẔ-**	**-ČG-**

The following six number roots are used when needed to designate numbers beyond ten when needed for counting and mathematical operations involving non-decimal number bases up to base-16. They may also be used as "short-cut" substitutes for the standard decimal/centesimal forms using the TNX affix.

11	12	13	14	15
-CG-	**-JD-**	**-L̦J-**	**-BC-**	**-ȚẔ-**

Whole numbers are full formatives signifying a set containing the particular number of members. The "simple" everyday counting system is base-100 (the mathematical sub-language will utilize base-12). Beginning with 'two', the Stem & Specification pattern is illustrated by the root -**Z**- 'three' below:

-**Z**- **'THREE / TRINARY'** Associated Affix: **3XX**			
	STEM 1	**STEM 2**	**STEM 3**
BSC	(to be a) set or group of three entities; (to be) a trio	(to be) something manifesting three aspects / facets; to manifest trinariness; be trinary	(to be) the third entity/party in a group or sequence
CTE	(to be) a party/entity of whom/which there are three	(to be) the state of having three aspects/facets; to be trinary; to be tri-fold or tri-faceted	(to be) the state of being third in a sequence/group/pattern
CSV	(to be) a process which determines/identifies a set as being three in number; to count out to three; to determine that there are three of something	(to be) a process which determines/identifies an entity as having three aspects/facets; identify/determine that something is trinary/tri-fold/tri-faceted	(to be) a process which determines/identifies an entity's sequential place in a sequence or group/pattern to be third
OBJ	(to be) one in a group or sequence of 3; to be one of 3	(to be) one of the aspects/facets of a trinary, tri-fold, tri-faceted entity	(to be) the entity/party whose numerical place in a sequence/group/pattern is third

Numbers from 11 through 99 are formed utilizing the TNX affix. Beginning with the number 101, numbers are formed as in Ithkuil-2011 using the COMITATIVE case and the COO affix. Having no multiples, the roots for 'ZERO' and 'ONE' have a different Stem & Specification pattern:

-VR- ‘ZERO / NULL’			
	STEM 1	**STEM 2**	**STEM 3**
BSC	(to be) zero as the empty-set / a set having no members; to have no quantity or amount	(to be) the zero-dimension; to have geometrically no length, area or volume	(to be) the baseline “zero”-state or null-state in a sequence, hierarchy, gradient, pattern, etc.
CTE	(to be) a party/entity of whom/which there are no members	(to be) the state of having no substance/tangibility due to being zero-dimensional	(to be) the state of being the baseline “zero”-state or null-state
CSV	(to be) a set having no members; to have no (i.e., zero) members in a set	(to be) the process/act of determining/identifying zero-dimensionality	(to be) a process which determines/identifies an entity’s being the baseline “zero”-state or null-state
OBJ	(to be) a null value / a value for a parameter that is undefined and/or for which the expected or standard value(s) is/are inapplicable	(to be) an entity having zero-dimensionality; (to be) a Euclidean point; to have geometrically no length, area or volume, i.e., to be a Euclidean point	(to be) the entity/party in the baseline “zero”-state or null-state in a sequence, hierarchy, gradient, pattern, etc.

-LL- ‘ONE / UNITY’			
	STEM 1	**STEM 2**	**STEM 3**
BSC	(to be) a set or group of one; to have one member	(to be) something (quasi-)indivisible, (quasi-)inseparable, unified, unitary, united, a union, a unit	(to be) the first entity/party in a group or sequence
CTE	(to be) a party/entity of whom/which there is only one	(to be) the state of having only one functional aspect/facet; to function/manifest as a unified whole or unit	(to be) the state of being first in a sequence/group/pattern
CSV	(to be) a process which determines/identifies a set as being one in number; to count out to one; to determine that there is only one of something	(to be) a process which determines/identifies an entity as having only one functional aspect/facet; to determine that an entity is a (quasi-)indivisible whole/unit	(to be) a process which determines/identifies an entity’s sequential place in a sequence or group/pattern to be first
OBJ	[same as CTE]	(to be) the party/entity having only one functional aspect/facet; to be an entity which functions/manifests as single unit	(to be) the entity/party whose numerical place in a sequence/group/pattern is first

-TF- ‘ADDITION / SUBTRACTION’			
	STEM 1	**STEM 2**	**STEM 3**
BSC	(to be) an act/instance of changing the number of something via addition or subtraction	(to be) an act/instance of adding the number of something; to add something	(to be) an act/instance of subtracting the number of something; to subtract something
CTE	(to be) the party/entity that performs/initiates an act of adding or subtracting	(to be) the party/entity that performs/initiates an act of addition	(to be) the party/entity that performs/initiates an act of subtraction
CSV	(to be) a process of adding or subtracting	(to be) a process of addition	(to be) a process of subtraction
OBJ	(to be) the party/entity whose number is changed via addition or subtraction	(to be) the party/entity whose number is added to	(to be) the party/entity whose number is subtracted

-ẔV- ‘MULTIPLICATION / DIVISION’			
	STEM 1	**STEM 2**	**STEM 3**
BSC	(to be) an act/instance of changing the number of something via multiplication or division; to multiply either by iteration or by division	(to be) an act/instance of multiplying the number of something; to multiply something	(to be) an act/instance of dividing the number of something; to divide something
CTE	(to be) the party/entity that performs/initiates an act of multiplying or dividing something	(to be) the party/entity that performs/initiates an act of multiplying something	(to be) the party/entity that performs/initiates an act of dividing something
CSV	(to be) a process of increasing via multiplication or division	(to be) a process of multiplication	(to be) a process of division
OBJ	(to be) the party/entity whose number is multiplied or divided	(to be) the party/entity whose number is multiplied	(to be) the party/entity whose number is divided

-NT- ‘COUNTING / TALLYING / SEQUENTIAL ENUMERATION’		Associated Affix: **SEQ**
BSC	**STEM 1:** (to be) a sequential numerical count(ing) of entities by natural numbers (i.e., positive integers) to determine the numerical quantity thereof; to count a set of entities, to tally the number of entities	**STEM 2:** (to be) a sequential numerical labeling [of entities by natural numbers] for purpose of identification, individuation, arrangement, etc. **STEM 3:** (to be) a numerical(ly-based) pattern/arrangement; to enumerate
CTE	(to be) the [natural] number of entities counted; the count; to count (up)/tally using natural numbers [= focusing on the accrual/addition of numbers in anticipation of the total/tally]	
CSV	(to be) the process of counting/tallying; to count (up)/tally [= focusing on the counting process itself irrespective of the eventual total]	
OBJ	(to be) the entity/entities so counted	

-NTH- 'NUMBER / INTEGER / RATIONAL NUMBER'		
BSC	**STEM 1:** (to be) a number; to express numerically	**STEM 2:** (to be) an integer, express via integers *
CTE	(to be) a state of being expressed/expressable numerically	**STEM 3:** (to be) a rational number, express via rational numbers * (i.e., expressable as a ratio of two integers)
CSV	(to be) an act of performing something by expressing it numerically, do something by using numbers	
OBJ	(to be) a numerical expression	

***** concatenate an appropriate stem from the root -**TVY**- with this stem to indicate positive or negative numbers *

-RNT- 'REAL NUMBER / IRRATIONAL NUMBER / IMAGINARY NUMBER'		
BSC	**STEM 1:** (to be) a real number	**STEM 2:** (to be) an irrational number (i.e., not expressable as a ratio of two integers)
CTE	(to be) a state of being expressed/expressable as a real number	**STEM 3:** (to be) an imaginary/complex number [i.e., a multiple of *i*]
CSV	(to be) an act of performing something by expressing it as a real number, do something by using real numbers	
OBJ	(to be) a real number expression	

-NTHW- 'VARIABLE / COEFFICIENT / CONSTANT'		
BSC	**STEM 1:** (to be) a mathematical variable; to express as a mathematical variable	**STEM 2:** (to be) a mathematical coefficient; to express as a mathematical coefficient
CTE	(to be) a state of being expressed/expressable as a mathematical variable	**STEM 3:** (to be) a mathematical constant; to express as a mathematical constant
CSV	(to be) an act of performing something by expressing it as a mathematical variable, do something by using a mathematical variable	
OBJ	(to be) a mathematical function, a mathematical expression using a variable or variables	

-ŇZX- ‘ABSTRACT OBJECT / ABSTRACT STRUCTURE / ABSTRACT OPERATION’			
BSC	**STEM 1:** (to be) an abstract object; to operate/ function as an abstract object [i.e., an abstract entity governed by abstract rules]	**STEM 2:** (to be) an abstract structure; to operate/function as an abstract structure [i.e., a grouping of abstract objects that displays certain predetermined qualities and supports certain operations, e.g., an algebraic structure]	**STEM 3:** (to be) an operation performed upon an abstract object or structure; to perform such an operation [i.e., a rule-based transformation of abstract objects]
CTE	(to be) a defining property (or set of defining properties) of an abstract object	(to be) a component of such a structure	(to be) the underlying rule that forms the basis for an abstract operation
CSV	(to be) the specific identity of an abstract object; to identify an abstract object	(to be) the defining purpose/function of such a structure; to define the purpose/function of such a structure	(to be) an act/process of performing/carrying out such an operation; to peform/carry out/apply an abstract operation
OBJ	(to be) a property/attribute peculiar to a particular abstract object	(to be) a property/operation manifested or supported by such a structure	(to be) the (desired) transformation precipitated by performance/application of an abstract operation

NOTE: Use an appropriate degree of the OAU affix with the stems of the above root to generate a term for “argument” or “abstract entity” meaning a logical participant or referenced entity within a logical argument, e.g., “In linguistics, we speak of the various nouns within a sentence as being *arguments* to a verb.”

6.4.2 Shapes and Forms

-FŘ- ‘SHAPE / FORM / FIGURE’		Associated Affix: **SHP**
BSC	**Stem 1:** (to be/manifest) a two-dimensional shape or outline form — [both the shape and the entity manifesting that shape]	**Stem 2:** (to be/manifest) a three-dimensional shape/form **Stem 3:** (to be a) figure (reminiscent of something) based on shape/form
CTE	(to be) the entity manifesting a particular 2-D shape or outline form	
CSV	(to be/manifest) a two-dimensional shape or outline form	
OBJ	(to be) the background or “negative space” behind a 2-dimensional shape or outline form	

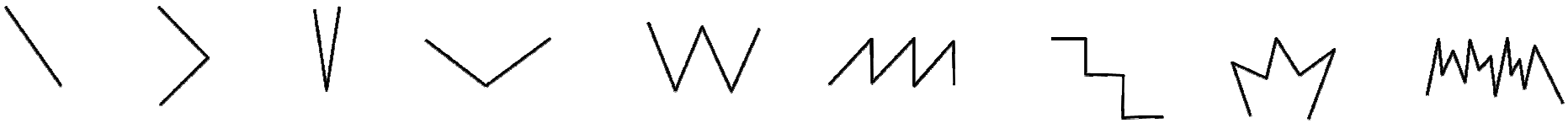
The Specification pattern of the above root is also used for the stems of the following root:

-ŘFŘ- ‘CONTOUR / OUTLINE / MOLD / CAST’
Stem 1: (to be/manifest a) two-dimensional contour, pattern, or outline as a representation/model for something
Stem 2: (to be/manifest a) three-dimensional contour, pattern, or outline as a representation/model for something
Stem 3: (to be a) mold/pattern/cast (of/for something)

-JB- **'TWO-DIMENSIONAL DECORATIVE PATTERN/TEXTURE'** Affix: **PAT**		
BSC	**STEM 1:** (to be/manifest) a two-dimensional (quasi-)overlying simple visual surface design/pattern (= repetitive visual design)— [both the design/pattern and the entity manifesting such]	**STEM 2:** (to be/manifest) a two-dimensional (quasi-) overlying complex/multi-faced visual surface pattern (= complex repetitive visual design) **STEM 3:** (to be/manifest a) two-dimensional (quasi-) overlying visual surface texture (= repetitive visual-tactile design)
CTE	(to be) something manifesting a 2-D overlying simple visual surface pattern	
CSV	(to be) a 2-D (quasi-)overlying simple visual surface pattern	
OBJ	(to be) the background upon or over which a 2-dimensional overlying design/pattern appears	

	-CKL- OPEN-ENDED LINEAR OUTLINE SHAPE	
BSC	**Stem 1:** (be) an open-ended (quasi-)linear outline shape (i.e. for describing the outline shape of any visual or tactile object/entity), e.g., an arc, a straight line, an S-shape, a V-shape, a squiggly line, etc.	**Stem 2:** (be) the (quasi-)planar extension (in 3-dimensional space) of the Stem 1 linear shape, e.g., a hemisphere shape based on the Stem-1 linear shape of an arc, or a notch- or wedge-shaped indentation based on the Stem-1 linear shape of a V-shape. **Stem 3:** (be) the "negative" space delineated by a (quasi-)linear outline shape, e.g., the pointed area of one's plane of vision created as the negative space set off by something in the foreground having a V-shaped outline.
CTE	(be) the state of having a particular (quasi-)linear outline shape	
CSV	(be) the act of discerning/observing/perceiving a (quasi-)linear outline shape	
OBJ	(be) an object/entity which manifests the particular (quasi-)linear outline shape	

The following roots all have a Stem & Specification pattern based on the root -**CKL**- above. The English names given each shape are, in many cases, rather stilted given the absence of concise nomenclature being available. Note also that the directional orientation of the shape as shown is arbitrary and irrelevant:



-VŘ- STRAIGHT LINE | **-PLW-** RIGHT-ANGLE | **-PLY-** ACUTE-ANGLE | **-FTY-** OBTUSE-ANGLE | **-JKL-** W-SHAPED | **-JKR-** SAWTOOTH-SHAPED | **-JKH-** STAIRCASE-SHAPED | **-VTÇ-** 3-POINTED SHAPE | **-KSMY-** IRREGULAR POINTS



-ḌGL- ARC | **-MKŘ-** SEMI-CIRCLE | **-NZŘ-** CIRCULAR | **-GBL-** HUMPED-SHAPE | **-ÇBŘ-** U-SHAPED | **-GBR-** DOUBLE-HUMPED | **-GZB-** S-SHAPED



-KṬM-
BULLET-SHAPED



-KṬK-
ARISTATE*



-KṬTḶ-
MUCRONATE*



-KṬN-
FLATTENED ARISTATE *



-JBL-
FLATTENED BULLET-SHAPED



-ḌDL-
SCALLOP-SHELL-SHAPED



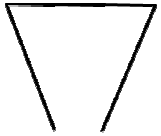
-ḌDW-
TRUNCATE*



-VGL-
SPADE-SHAPED



-ŽTY-
POINTED SPADE-SHAPED



-XPY-
WEDGE-SHAPED



-XPR-
CONCAVE WEDGE



-ẔGL-
Y-SHAPED



-MNY-
TRIPLE-LOBE



-SSKY-
IRREGULAR LINEAR



-PFW-
TOOTH-SHAPED



-PFY-
AURICULATE*



-PFČ-
OBCORDATE *



-PFR-
PAIRED LOBES



-ŘFL-
ROUNDED Y-SHAPED



-ŘFY-
HASTATE *



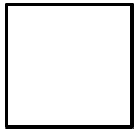
-ŘTŘ-
MIXED ROUNDED & POINTED

* Term borrowed from biology/botany for a particular leaf shape.

-CKR- 'CLOSED LINEAR OUTLINE SHAPE'		
BSC	**Stem 1:** (be) a closed (quasi-)linear outline shape (i.e. for describing the outline shape of any visual or tactile object/entity	**Stem 2:** (be) a (quasi-)planar object/entity whose edge(s) constitute the Stem 1 closed linear shape, e.g., a cookie in the shape of a circle, or a stop-sign in the shape of an octagon. **Stem 3:** (be) the background "negative" space delineated by a closed (quasi-)linear outline shape, e.g., the area of one's plane of vision with a "hole" in it, created as the negative space set off by something in the foreground having a closed-linear outline.
CTE	(be) the state of having a particular closed (quasi-)linear outline shape	
CSV	(be) the act of discerning/observing/perceiving a closed (quasi-)linear outline shape	
OBJ	(be) an object/entity which manifests the particular closed (quasi-)linear outline shape	

The following roots all have a Stem & Specification pattern based on the above root.

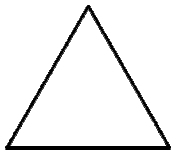
In general, the pattern of roots for planar and three-dimensional shapes is as follows: The basic closed planar shape (e.g., a circle, a square, an equilateral triangle, etc.) has a tri-consonantal root ending in -**L**. The three-dimensional extension of such a root with a rectilinear base changes this -**L** to -**R**. The three-dimensional extension of such a root with a radial or circular base changes it to -**W**. If the shape has an associated open-ended linear outline form, this is expressed by changing the last consonant of the root to -**Ř** or -**Y**.



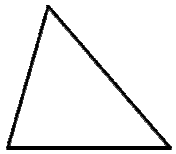
-PŠL-
SQUARE



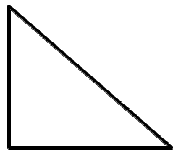
-KŠL-
RECTANGLE



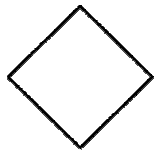
-ZL-
ISOSCELES TRIANGLE



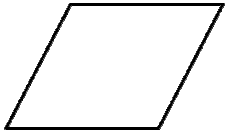
-ŘZL-
SCALENE TRIANGLE



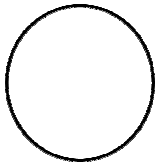
-RZL-
RIGHT TRIANGLE



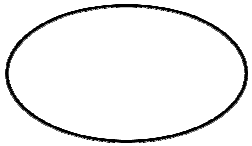
-RPŠL-
DIAMOND SHAPED



-ŘPŠL-
RHOMBOID



-NZL-
CIRCLE



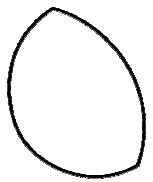
-MZL-
ELLIPSE



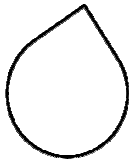
-KKL-
SLIVER-MOON SHAPED



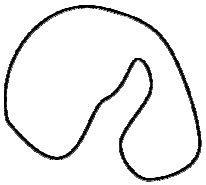
-MKL-
HALF-MOON SHAPED



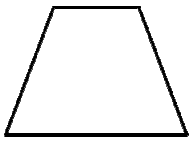
-KKR-
GIBBOUS-MOON SHAPED



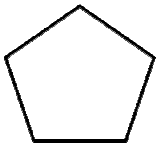
-SBL-
TEARDROP-SHAPED



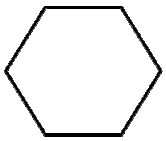
-SSKL-
IRREGULAR ROUNDED SHAPE



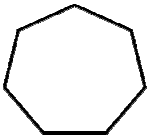
-LPŠL-
TRAPEZOID



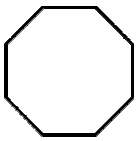
-STL-
PENTAGON



-CPL-
HEXAGON



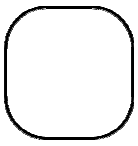
-NSL-
HEPTAGON



-ČKL-
OCTAGON



-
DO
TRU
SO



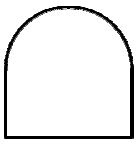
-NZM-
ROUNDED SQUARE



-LKŠL-
ROUNDED RECTANGLE



-NZVL-
OVAL



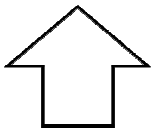
-ÇBL-
HALF-ROUNDED SQUARE



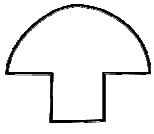
-TBL-
THICK ARCH SHAPED



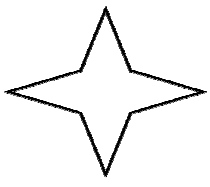
VA
S



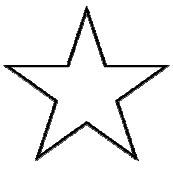
-VKL-
ARROW-SHAPED



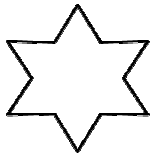
-BŽV-
MUSHROOM-SHAPED



-PŠŘ-
4-POINTED STAR



-STŘ-
5-POINTED STAR



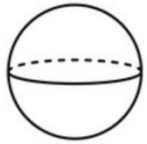
-CPŘ-
SIX-POINTED STAR



-SI
MULTI-
ST

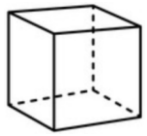
-TĻK-	‘THREE-DIMENSIONAL FORM/LATTICE’	
BSC	**Stem 1:** (be) a solid three-dimensional outline form/shape, as determined by the holistic structure of its planar surfaces (e.g., a cube, a sphere, a pyramidal shape, etc.)	**Stem 2:** [same as stem 1 except that the 3-dimensional form is hollow, consisting only of the collection of surface planes or geodesic/curved plane] **Stem 3:** (be) a a three-dimensional lattice form/shape, as determined by the collective structure/lattice formed by its linear edges (e.g., a cube-shaped lattice formed from eight linear elements of equal length whose end-points are connected to two others at right angles, a lattice in the form of a sphere, a pyramid-shaped lattice, etc.)
CTE	(be) the state of having a particular three-dimensional outline form/shape	
CSV	(be) the act of discerning/observing/perceiving a three-dimensional outline form/shape	
OBJ	(be) an object/entity which manifests the particular three-dimensional outline form/shape	

The following roots have the same Stem/Specification structure as the root above:



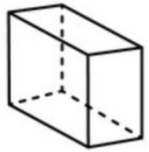
-NZW-

SPHERE



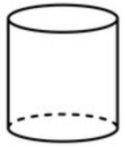
-PŠR-

CUBE



-KŠR-

RECTANGULAR CUBOID



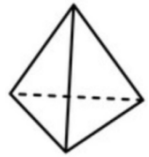
-NZN-

CYLINDER



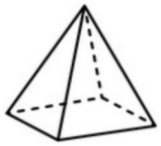
-NZY-

CONE



-PŠN-

TETRAHEDRON



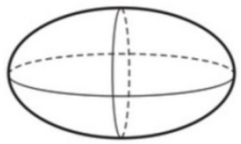
-ZR-

PYRAMID



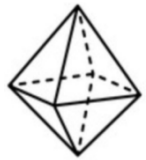
-MKW-

HEMISPHERE



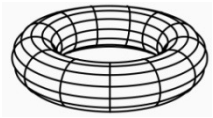
-MZW-

ELLIPSOID



-RČKW-

TRIANGULAR OCTAHEDRON



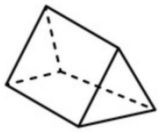
-TBW-

TORUS



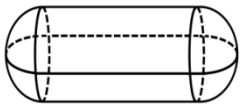
-TBY-

HALF-TORUS



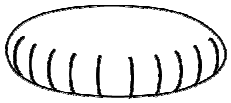
-RZR-

EQUILATERAL TRIANGULAR PRISM SHAPE



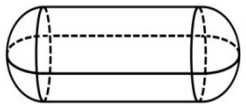
-NZVW-

OVOID



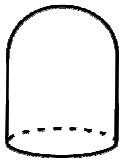
-NZR-

DISC



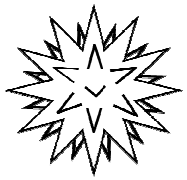
-MZR-

LOZENGE SHAPE [i.e., same as ovoid except that width-wise cross-section is an oval or ellipse rather than a circle]



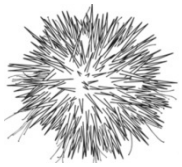
-ÇBW-

ROUND-TOPPED CYLINDER



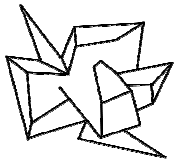
-SKR-

3-DIMENSIONAL MULTI-POINTED STAR



-VKF-

3-DIMENSIONAL SPIKED SPHEROID



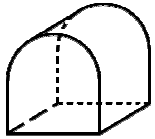
-KSMR-

3-DIMENSIONAL IRREGULAR POLYHEDRON



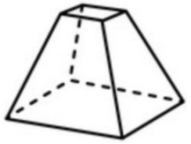
-SSKR-

3-DIMENSIONAL IRREGULAR "AMOEBOID" SHAPE



-ÇBR-

ROUND-TOPPED CUBOID



-LPŠR-

TRUNCATED PYRAMID

Additional Roots:

-**ŇBL**- polygon [for a polygon with a specific number of sides more than eight, concatenate a number stem with either COMPOSITIVE format or UTILITATIVE format into this root]

-**ŠBL**- polyhedron [for a polyhedron with a specific number of faces more than eight, concatenate a number stem with either COMPOSITIVE format or UTILITATIVE format into this root]

-**RNZM**- concave disc (i.e., shaped like a Life-Saver piece of candy)

-**RNZN**- half-concave disc (i.e., concave lens-shaped)

-**RNZW**- convex disc (i.e., convex lens-shaped)

The following affix is available to modify the above stems for 3-dimensional forms:

-jk	VPF Variance From Prototypical 3-D Form
1	long axis obliquely angled/inclined away from the vertical/right-angled/rectilinear/perpendicular
2	elongated lengthwise/heightwise
3	elongated widthwise/girthwise
4	curved away from right-angled/rectilinear orientation
5	truncated at small/pointed/shorter/smaller end at angle parallel to base
6	truncated at small/pointed/shorter/smaller end at oblique angle to base
7	a quasi-planar longitudinal (i.e., perpendicular to base) slice through 3-D form X
8	a quasi-planar oblique slice through 3-D form/shape X
9	a quasi-planar latitudinal (i.e., parallel to base) slice through 3-D form X

-ļř	TDP Three-Dimensional Polyhedral Forms
1	symmetrical 3-dimensional polyhedron having X-number of triangular faces
2	symmetrical 3-dimensional polyhedron having X-number of quadrilateral faces
3	symmetrical 3-dimensional polyhedron having X-number of mixed triangular and quadrilateral faces
4	symmetrical 3-dimensional polyhedron having X-number of mixed polygonal faces
5	a 3-dimensional form/object whose 2-dimensional/planar outline shape (i.e., against a background) is X
6	asymmetrical 3-dimensional polyhedron having X-number of mixed polygonal faces
7	asymmetrical 3-dimensional polyhedron having X-number of mixed triangular and quadrilateral faces
8	asymmetrical polygon having X-number of quadrilateral faces
9	asymetrical polygon having X-number of triangular faces

NOTE: For the roots -**ZR**-, -**RZR**-, and -**LPŠR**- above, concatenate a number-root to create words for X-sided pyramidal forms or X-sided prism forms.

-ļn	CNW Conway Polyhedral Mathematical Operations
1	dual
2	truncate
3	subdivide
4	augment
5	kis
6	bevel
7	chamfer
8	join
9	ambo

6.4.3 Chemical Elements, Compounds and Substances

-RẔ- 'CHEMICAL ELEMENT'		
BSC	**Stem 1:** (to be) a chemical element	**Stem 2:** (to be) a positive ion of a chemical element **Stem 3:** (to be) a negative ion of a chemical element
CTE	(to be) an elemental state; at a level involving individual chemical elements	
CSV	(to be) a state/act of manipulating/combining chemical elements; to perform chemistry	
OBJ	(to be) a chemical element in its naturally occurring (homonuclear) state, e.g., O_2	

CPT version of the above 3 stems refers respectively to the oxidation state of zero, oxidized, and reduced ions.

The roots for the elements below follow the same Specification and Stem pattern as the root -**RẔ**- 'CHEMICAL ELEMENT' above:

-PÇK-	Hydrogen	**-RTŇ-**	Vanadium	**-ẔTW-**	Cobalt	**-ŠPW-**	Boron	**-ŇN-**	Oxygen
-PÇKW-	Lithium	**-RTŇW-**	Niobium	**-ẔTY-**	Rhodium	**-ŠPY-**	Aluminum	**-ŇZD-**	Sulfur
-PÇKY-	Sodium	**-RSŇ-**	Tantalum	**-ẔTF-**	Iridium	**-ŠPL-**	Gallium	**-ŇZDW-**	Selenium
-PÇKL-	Potassium	**-RSŇW-**	Dubnium	**-ẔTV-**	Meitnerium	**-ŠPR-**	Indium	**-ŇZDY-**	Tellurium
-PÇKR-	Rubidium					**-ŠPŘ-**	Thallium	**-ŇZDL-**	Polonium
-PÇKŘ-	Caesium	**-ŠKR-**	Chromium	**-XDW-**	Nickel	**-ŠPŠ-**	Nihonium	**-ŇZDR-**	Livermorium
-PÇKF-	Francium	**-ŠKW-**	Molybdenum	**-XDY-**	Palladium				
		-ŠKY-	Tungsten	**-XDL-**	Platinum	**-ŽPL-**	Carbon	**-BZTW-**	Fluorine
-LVG-	Beryllium	**-ŠKŘ-**	Seaborgium	**-XDR-**	Darmstadtium	**-ŽPW-**	Silicon	**-BZTY-**	Chlorine
-LVGW-	Magnesium					**-ŽPY-**	Germanium	**-BZTL-**	Bromine
-LVGY-	Calcium	**-TĻN-**	Manganese	**-MẔ-**	Copper	**-ŽPR-**	Tin	**-BZTR-**	Iodine
-LVGL-	Strontium	**-TĻNW-**	Technetium	**-RĮ-**	Silver	**-ŽPŘ-**	Lead	**-BZTŘ-**	Astatine
-LVGR-	Barium	**-TĻŇ-**	Rhenium	**-LJ-**	Gold	**-ŽPF-**	Flerovium	**-BZTĻ-**	Tennessine
-LVGŘ-	Radium	**-TĻŇW-**	Bohrium	**-LJŇW-**	Roentgenium				
-LVGV-	Scandium					**-LXT-**	Nitrogen	**-GZPW-**	Helium
-LVGZ-	Yttrium	**-SGL-**	Iron	**-ĐBW-**	Zinc	**-LXTW-**	Phosphorus	**-GZPY-**	Neon
		-SGW-	Ruthenium	**-ĐBY-**	Cadmium	**-LXTY-**	Arsenic	**-GZPL-**	Argon
-PĻW-	Titanium	**-SGY-**	Osmium	**-ĐBL-**	Mercury	**-LXTL-**	Antimony	**-GZPR-**	Krypton
-PĻM-	Zirconium	**-SGV-**	Hassium	**-ĐBR-**	Copernicium	**-LXTR-**	Bismuth	**-GZPŘ-**	Xenon
-PĻN-	Hafnium					**-LXTŘ-**	Moscovium	**-GZPF-**	Radon
-PĻŇ-	Rutherfordium							**-GZPĻ-**	Oganesson

-ŘCP-	Lanthanum	**-ŘCPR-**	Terbium	**-ŘPST-**	Actinium	**-ŘPSC-**	Berkelium
-ŘCPW-	Cerium	**-ŘCPS-**	Dysprosium	**-ŘPSTW-**	Thorium	**-ŘPSČ-**	Californium
-ŘCPY-	Praseodymium	**-ŘCPŠ-**	Holmium	**-ŘPSTY-**	Protactinium	**-ŘPSCW-**	Einsteinium
-ŘCPÇ-	Neodymium	**-ŘCC-**	Erbium	**-ŘPSTL-**	Uranium	**-ŘPSČW-**	Fermium
-ŘCPL-	Promethium	**-ŘCCW-**	Thulium	**-ŘPSTR-**	Neptunium	**-ŘPSS-**	Mendelevium
-ŘCPF-	Samarium	**-ŘCČ-**	Ytterbium	**-ŘPSTŘ-**	Plutonium	**-ŘPSSW-**	Nobelium
-ŘCPĻ-	Europium	**-ŘCČW-**	Lutetium	**-ŘPSTF-**	Americium	**-ŘPSSY-**	Lawrencium
-ŘCPŘ-	Gadolinium			**-ŘPSTĻ-**	Curium		

The following new affixes are used for naming chemical compounds:

Ionic Compounds — inorganic compounds between metals [cations] and non-metals [anions] (like English **-ide** suffix but with electron charge explicitly stated):

-ltř	**ION Ionic Compounding Elements**
1	transition metal cation with +1 charge
2	transition metal cation with +2 charge
3	transition metal cation with +3 charge
4	transition metal cation with +4 charge
5	hydrogen + anion acid (i.e., hydro-...-ic acid), e.g., when used w/ chlorine, this results in hydrochloric acid, i.e., hydrochloride / chlorhydrate
6	compounding anion with -4 charge
7	compounding anion with -3 charge
8	compounding anion with -2 charge
9	compounding anion with -1 charge

Molecular Compounds — inorganic compounds between nonmetals:

-rtl	**AMC Number of Atoms in a Molecular Compound**
1	deca- *
2	di-
3	tri-
4	tetra-
5	penta-
6	hexa-
7	hepta-
8	octa-
9	nona-

* A single atom (equivalent to English “mono-“) is indicated by the absence of the suffix; therefore Degree 1 of this affix is used to signify ten atoms (deca-).

The following four affixes provide shortcut means of naming some common polyatomic ions:

-cţ	PIA Polyatomic Ions A
1	sulfate
2	nitrate
3	chlorate
4	arsenate
5	carbonate
6	chromate
7	bromate
8	phosphate
9	iodate

-ẓţ	PIB Polyatomic Ions B
1	sulfite
2	nitrite
3	chlorite
4	arsenite
5	hydroxide
6	chromite
7	bromite
8	phosphite
9	iodite

-čţ	PIC Polyatomic Ions C
1	peroxide
2	silicate
3	citrate
4	acetate
5	oxide
6	cyanide
7	tartrate
8	formate
9	oxalate

-jţ	PID Polyatomic Ions D
1	thiocyanate
2	thiosulfate
3	glycol
4	hexacyanoferrate
5	permanganate
6	succinate
7	azanide / amide
8	manganate
9	benzoate

-zţ	CPI Polyatomic Cations
1	ammonium
2	fluoronium
3	hydronium
4	pyrylium
5	phosphonium
6	guanidinium
7	tropylium
8	triphenylcarbenium
9	cyclopropenium

-žţ	PIC Additional Polyatomic Ionic Configurations
1	-ate (base oxyanion), e.g., sulfate
2	-ite (i.e., one less oxyanion than base form), e.g., sulfite
3	hypo- ... -ite (two less oxyanions than base form), e.g., hyposulfite
4	per- ... -ate (one more oxyanion than base form), e.g., persulfate
5	-ide (anionic element where electron charge not specified), e.g., sulfide
6	bi- ... -ate / hydrogen ... -ate (hydrogen + one oxyanion more than base form), e.g., bisulfate
7	dihydrogen ... -ate (two hydrogens + one oxyanion more than base form), e.g., dihydrogen phosphate
8	di- ... -ate (two atoms of an element + one oxyanion more than base form), e.g., dichromate
9	di ... -ide (two atoms of an element + base oxyanion), e.g., disulfide

The following nine affixes provide shortcut means of naming some functional groups for organic compounds:

-sḍ	GPA Functional Group A
1	alkane, alkyl-, -ane
2	alkene, alkenyl-, -ene
3	alkyne, alkynyl-, -yne
4	benzene deriv., phenyl-, benzene
5	halo-, alkyl halide
6	fluoro-, alkyl fluoride
7	chloro-, alkyl chloride
8	bromo-, alkyl bromide
9	iodo-, alkyl iodide

-sḷ	GPB Functional Group B
1	hydroxil, hydroxy-, -ol
2	carbonyl, oxo-, -oyl-, -one
3	aldehyde, -formyl-, -al
4	haloformyl, carbono...oyl-, -oyl halide
5	carbonate ester, alkoxycarbonyloxy-, alkyl carbonate
6	carboxylate, carboxy-, -oate
7	carboxyl, carboxy-, -oic acid
8	carboalcoxy, alkanoyloxy-, alkyl alkanoate
9	methoxy, methoxy-

-šḍ	GPC Functional Group C
1	hydroperoxy-, alkyl hydroperoxide
2	peroxy-, alkyl peroxide
3	ether, alkoxy-, alkyl ether
4	hemiacetal, alkoxy -ol, -al alkyl hemiacetal
5	hemiketal, alxoxy -ol, -one alkyl hemiketal
6	acetal, dialkoxy-, -al dialkyl acetal
7	ketal, dialcoxy-, -one dialkyl ketal
8	orthoester, -trialkoxy
9	orthocarbonate ester, tetralkoxy-, tetraalkyl orthocarbonate

-šḷ	GPD Functional Group D
1	methlenedioxy-, -dioxole
2	carboxylic anhydride, anhydride
3	carboxamide, carboxamido-, carbamoyl-, -amide
4	primary amine, amino-, -amine
5	secondary amine, amino-, -amine
6	tertiary amine, amino-, -amine
7	ammonio-, -ammonium
8	imide, imido-, -imide
9	azide, azido-, alkyl azide

-zḍ	GPE Functional Group E
1	primary ketimine, imino-, imine
2	secondary ketimine, imino-, -imine
3	primary aldimine, imino-, imine
4	secondary aldimine, imino-, -imine
5	azo diimide, azo-, -diazene
6	cyanate, cyanato-, alkyl cyanate
7	isocyanate, isocyanato-, alkyl isocyanate
8	nitrate, nitrooxy-, nitroxy-, alkyl nitrate
9	nitrite, nitrosooxy-, alkyl nitrite

-zḷ	GPF Functional Group F
1	nitrile, cyano-, alkanenitrile, alkyl cyanide
2	isonitrile, isocyano-, alkaneisonitrile, alkyl isocyanide
3	nitro compound, nitro-
4	nitroso compound, nitroso-, nitrosyl-
5	oxime
6	pyridyl, 4-pyridyl, 3-pyridyl, 2-pyridyl, -pyridine
7	carbamate, -carbamoyloxy-, -carbamate
8	phosphine, phosphanyl-, -phosphane
9	phosphonic acid, phosphono-, -phosphonic acid

-žḍ	GPG Functional Group G
1	phosphate, phosphonooxy-, O-phospono-, ... phosphate
2	phosphodiester, hydroxyphosphoryloxy-, di...hydrogen phosphate
3	boronic acid, borono-, ... boronic acid
4	boronate, O-alkylboronyl-, ... boronic acid di... ester
5	brorinic acid, hydroxyborino-, di... obrinic acid
6	borinate, O-alkoxydialkylboronyl-, di... borinic acid ... ester
7	alkylithium, -lithium
8	alylmagnesium halide, -magnesium halide
9	alkylaluminium, -aluminium / -aluminum

-žḷ	GPH Functional Group H
1	sulfhydryl, sulfanyl-, -thiol
2	sulfide, -sulfanyl-, di... sulfide
3	disulfide, -disulfanyl-, di... disulfide
4	sulfoxide, -sulfinyl-, di... sulfoxide
5	sulfone, -sulfonyl-, di... sulfone
6	sulfinic acid, sulfino-, -sulfinic acid
7	sulfonic acid, sulfo-, -sulfonic acid
8	sulfonate ester, alkoxysulfonyl-, -sulfonate
9	carbonothioyl, -thioyl-, sufanylidene-, -thione

-ẓḍ	GPJ Functional Group J
1	thiocyanate, thocyanato-, ...thiocyanate
2	isothiocyanate, isothiocyanato-, ...isothiocyanate
3	carbonothioyl, methanethioyl-, sufanylidene-, -thial
4	carbothioic S-acid, mercaptocarbonyl-, -thioic S-acid
5	carbothioic O-acid, hydroxythiocarbonyl-, -thioic O-acid
6	thiolester, S-alkyl-alkane-thioate
7	thionoester, O-alkyl-alkane-thioate
8	carbodithioic acid, dithiocarboxy-, -dithioic acid
9	dithiocarboxylic acid ester, -dithioate

-RẒB- 'CHEMICAL COMPOUND / CHEMICAL REACTION'		
BSC	**Stem 1:** (to be) a chemical compound; to consist of a chemical compound	**Stem 2:** (to be) a chemical reaction; to chemically react, undergo a chemical reaction **Stem 3:** (to be) an amount of energy released as a result of a chemical reaction; to release energy as a result of a chemical reaction
CTE	(to be) an instance (i.e.g, piece, drop, granule, plume, glob) of the substance comprising a chemical compound	
CSV	(to be) a chemical reaction resulting in the formation of a chemical compound; to chemically react, make/undergo a chemical reaction	
OBJ	(to be) a chemical compound in its naturally occurring state	

NOTE: for the meaning of the old FORMAL stems previously associated with this root, use the following new root:

- RẒBW- 'CHEMICAL COMPOUND BY TYPE OF MOLECULAR BOND'		
BSC	**Stem 1:** (to be) a chemical compound held together by covalent molecular bonds	**Stem 2:** (to be) a chemical compound held together by ionic bonds **Stem 3:** (to be) an intermetallic chemical compound or complex held together by metallic or coordinate covalent bonds
CTE	(to be) an instance (i.e.g, amount, piece, drop, granule, flume, glob) of the substance comprising a chemical compound held together by covalent molecular bonds	
CSV	(to be) a chemical reaction resulting in the formation of a chemical compound held together by covalent molecular bonds; the chemically react, make/undergo a chemical reaction involving covalent molecular bonds	
OBJ	(to be) a chemical compound held together by covalent molecular bonds in its naturally occurring state	

The following roots for common chemical compounds follow the same Specification pattern as the root **-RẔB-** above:

-FD-	table salt / rock salt (sodium chloride)
-TPF-	plaster of Paris (calcium sulfate hemi-hydrate)
-PPF-	talc / talcum
-DPF-	sodium bicarbonate (i.e., baking soda) $NaHCO_3$
-ŇPF-	washing soda (sodium carbonate decahydrate)
-RPF-	magnesia (magnesium hydroxide)
-ŘPF-	lye / soda lye / caustic soda (sodium hydroxide)
-XPF-	lime / quicklime (calcium oxide)
-ĻPF-	slaked lime / caustic lime (calcium hydroxide)
-ŠPF-	potash (potassium carbonate)
-GPF-	saltpeter (potassium nitrate)
-VPF-	bleaching powder (calcium hypochlorite)
-CPF-	borax (sodium tetraborate)
-ČPF-	emery / corundite
-ĐPF-	epsom salt (magnesium sulfate heptahydrate)
-KÇPF-	calcium oxychloride (i.e., bleaching powder)
-NĻPF-	caustic potash (potassium hydroxide)
-KŢPF-	silica
-DMY-	bleach (sodium hypochlorite)
-PMY-	zinc white / zinc oxide
-ZDW-	fluorspar / fluorite
-DNW-	carbolic acid (phenol)
-ZDY-	copper sulphate (i.e., blue vitriol)
-CBL-	carbon dioxide
-GVBL-	laughing gas (nitrous oxide)
-GVBR-	methane

-ẔTÇ-	dynamite
-XBW-	aspirin / ASA
-JNY-	calomel (mercurous chloride)
-KSPR-	gasoline
-KSPL-	kerosene
-KSPW-	benzene
-KSPY-	butane
-KSPŘ-	propane
-VML-	methanol / methyl alcohol / wood alcohol
-VMY-	grain alcohol (ethyl alcohol)
-TĻMW-	graphite
-TĻMY-	graphene
-LCM-	table sugar (sucrose)
-LCN-	glucose
-LCMW-	fructose
-LCMY-	dextrose
-LCNY-	sugar (any molecular “-ose” compound)
-ZDŘ-	asbestos
-XBŘ-	milk of magnesia
-KFTW-	acetone
-KFTY-	ammonia
-KFTL-	rubbing alcohol / isopropyl alcohol
-KFTR-	vinegar (acetic acid)
-KFTŘ-	glycerin (trihydroxy propanol)
-KFTÇ-	oil of vitriol (sulfuric acid)
-KFTH-	formalin / aqueous formaldehyde solution
-KFTHW-	muriatic acid (hydrochloric acid)

The following is a list of chemical compounds associated with organic lifeforms such as nucleic acids, essential nutrients, hormones, etc.

-**PFK**-	vitamin
-**PFKW**-	vitamin A / retinol / beta-carotene
-**PFKY**-	vitamin B_1 / thiamine
-**PFKL**-	vitamin B_2 / riboflavin
-**PFKR**-	vitamin B_3 / niacin / niacinamide
-**PFKŘ**-	vitamin B_5 / pantothenic acid
-**PFKÇ**-	vitamin B_6 / pyridoxine
-**PFKF**-	vitamin B_7 / biotin
-**PFKŢ**-	vitamin B_9 / folate / folic acid
-**PFKM**-	vitamin B_{12} / cobalamin
-**PFG**-	vitamin C / ascorbic acid
-**PFGW**-	vitamin D_2 / ergocalciferol
-**PFGY**-	vitamin D_3 / cholecalciferol
-**PFGL**-	vitamin E / tocopherol / tocotrienol
-**PFGR**-	vitamin K / phylloquinone /menaquinone
-**PFX**-	omega-3 fatty acid
-**PFXW**-	omega-6 fatty acid
-**PFXR**-	omega-9 fatty acid
-**PÇP**-	protein
-**PÇPW**-	nucleic acid
-**PÇPY**-	ribonucleic acid
-**PÇPL**-	deoxyribonucleic acid
-**PÇPR**-	nucleoside
-**PÇPŘ**-	nucleotide
-**PÇPÇ**-	adenine
-**PÇPF**-	cytosine
-**PÇPŢ**-	guanine
-**PÇPĻ**-	thymine
-**PÇPH**-	uracil
-**ŇZVL**-	peptide
-**ŇZVR**-	eicosanoid
-**ŇŽP**-	hormone
-**ŇŽPW**-	adrenaline
-**ŇŽPY**-	melatonin
-**ŇŽPL**-	norepinephrine
-**ŇŽPR**-	triiodothyronine
-**ŇŽPŘ**-	thyroxine
-**ŇŽPÇ**-	dopamine
-**ŇŽPF**-	prostaglandin
-**ŇŽPŢ**-	leukotriene
-**ŇŽPĻ**-	prostacyclin
-**ŇŽPH**-	thromboxane
-**ŇTP**-	amylin
-**ŇTPW**-	anti-Müllerian hormone
-**ŇTPY**-	adiponectin
-**ŇTPL**-	corticotropin
-**ŇTPR**-	angiotensin
-**ŇTPŘ**-	angiotensinogen
-**ŇTPÇ**-	vasopressin
-**ŇTPF**-	brain natriuretic peptide
-**ŇTPŢ**-	calcitonin
-**ŇTPĻ**-	cholecystokinin
-**ŇTPH**-	corticotropin-releasing hormone
-**ŇFP**-	cortistatin
-**ŇFPW**-	enkephalin
-**ŇFPY**-	endothelin
-**ŇFPL**-	erythropoietin
-**ŇFPR**-	follicle-stimulating hormone
-**MSG**-	androgen
-**MSGW**-	testosterone
-**MSGY**-	dehydroepiandrosterone
-**MSGL**-	androstenedione
-**MSGR**-	dihydrotestosterone
-**MSB**-	estrogen
-**MSBW**-	estradiol
-**MSBY**-	estrone
-**MSBL**-	estriol
-**MSBR**-	progestogen
-**MSBŘ**-	progesterone
-**MSM**-	corticoid
-**MSMW**-	aldosterone
-**MSMY**-	cortisol
-**MSN**-	steroid
-**MSNW**-	secosteroid
-**MSNY**-	calcitriol
-**MSŇ**-	calcidiol
-**MSD**-	thyroglobulin
-**MSDW**-	troponin I
-**MSDY**-	troponin T
-**MSDL**-	creatine kinase
-**MSDR**-	prostate specific antigen
-**MSDŘ**-	carcinoembryonic antigen
-**MFB**-	alpha fetoprotein
-**MFBW**-	myoglobin
-**MFBY**-	bilirubin
-**MFBL**-	hemoglobin
-**MFBR**-	ceruloplasmin

-**PÇŇ**- ferritin
-**PÇŇW**- C-reactive protein
-**PÇB**- amino acid
-**PÇBW**- histidine
-**PÇBY**- isoleucine
-**PÇBL**- leucine
-**PÇBR**- lysine
-**PÇBŘ**- methionine
-**PÇG**- phenylalanine
-**PÇGW**- threonine
-**PÇGY**- tryptophan
-**PÇGL**- valine
-**PÇGR**- arginine
-**PÇGŘ**- cystine
-**PÇD**- glutamine
-**PÇDW**- glycine
-**PÇDY**- proline
-**PÇDL**- tyrosine
-**PÇDR**- alanine
-**PÇDŘ**- aspartic acid
-**PÇF**- asparagine
-**PÇFW**- glutamic acid
-**PÇFY**- serine
-**PÇFL**- selenocysteine
-**PÇFR**- pyrrolysine
-**PÇFŘ**- taurine
-**PÇTW**- choline
-**PÇTY**- inositol
-**PÇTL**- polyphenol / lignan
-**PÇTR**- flavonoid

-**ŇFPŘ**- galanin
-**ŇFPÇ**- gastric inhibitory polypeptide
-**ŇFPF**- gastrin
-**ŇFPŢ**- ghrelin
-**ŇFPH**- glucagon
-**ŇFK**- glucagon-like peptide 1
-**ŇFKW**- gonadotropin-releasing hormone
-**ŇFKY**- hepcidin
-**ŇFKL**- human chorionic gonadotropin
-**ŇFKR**- growth hormone
-**ŇFKŘ**- inhibin
-**ŇFKÇ**- insulin
-**ŇFKF**- somatomedin
-**ŇFKŢ**- leptin
-**ŇFKM**- lipotropin
-**ŇFKN**- luteinizing hormone
-**ŇFKS**- melanocyte stimulating hormone
-**ŇFKŠ**- motilin
-**ŇFKH**- orexin
-**ŇŢP**- osteocalcin
-**ŇŢPW**- oxytocin / pitocin
-**ŇŢPY**- pancreatic polypeptide
-**ŇŢPL**- parathyroid hormone
-**ŇŢPR**- pituitary adenylate cyclase-activating peptide
-**ŇŢPŘ**- prolactin
-**ŇŢPÇ**- prolactin-releasing hormone
-**ŇŢPF**- relaxin
-**ŇŢPŢ**- renin
-**ŇŢPĻ**- secretin
-**ŇŢPH**- somatostatin

-**MFG**- pyruvate
-**MFGW**- uric acid
-**MFGY**- urea
-**MFGL**- lactate
-**MFGR**- creatinine
-**MFGŘ**- haptoglobin
-**MFD**- cholesterol
-**MFDW**- high-density lipoprotein
-**MFDY**- low-density lipoprotein
-**MFDL**- triglyceride
-**MFDR**- alpha-1 antitrypsin
-**MFDŘ**- transferrin
-**MSV**- fibrinogem
-**MSVW**- albumin
-**MSVY**- enzyme

-**PÇTŘ**- resveratrol

-**ŇŢF**- thrombopoietin
-**ŇŢFW**- thyroid-stimulating hormone
-**ŇŢFY**- vasoactive intestinal peptide
-**ŇŢFL**- guanylin
-**ŇŢFR**- uroguanylin
-**ŇŢFŘ**- ventricular natriuretic peptide

-SY- **'SUBSTANCE / MATERIAL / WHAT SOMETHING CONSISTS OF OR IS COMPOSED/MADE OF'**		
BSC	**STEM 1:** (to be a) an instance/amount of a material substance; (to be a) manifestation of matter/material/ "stuff" / something material — [both the substance itself and the form/entity consisting thereof]	**STEM 2:** (to be) a state of consisting of or being composed/made of a particular material substance; to consist of/be composed of/made (out) of a particular material/substance **STEM 3:** (to be/manifest a) material consistency, i.e., a discernible (quasi-)tactile or (quasi-)visual or other (quasi-)sensory attribute/quality/characteristic)
CTE	(to be) the material substance of which something consists or is made; something material	
CSV	(to be) a form taken by something material, the shape/form of something material	
OBJ	(to be) an object/entity made from or consisting of a particular material	

The following roots for common substances/materials follow the same Specification pattern as the stems of the above root **-SY- :**

-**NDW**- dirt/earth/soil
-**NDM**- clay
-**NTF**- sand
-**NTFW**- lava
-**NTÇ**- tar/pitch
-**NTW**- oil (plant-derived)
-**NTY**- petroleum
-**NTŘ**- latex (natural)
-**NDR**- wood
-**MNW**- honey
-**MTÇ**- wax
-**LTV**- silk
-**LTH**- ivory
-**PNW**- starch (amylose/amylopectin)

-**TĻM**- plastic
-**KSY**- metal
-**KSV**- steel / metal alloy
-**XTW**- ceramic
-**XTY**- porcelain
-**XMW**- plaster
-**XMY**- mortar
-**XNW**- adobe
-**XNY**- cement/concrete
-**BXW**- brick
-**TTW**- glass
-**TTY**- enamel
-**TTŘ**- tile
-**ŽTÇ**- asphalt/bitumen

-**ŇŽK**- polystyrene
-**ŇŽKW**- polystyrene foam ("styrofoam")
-**ŇŽKY**- polyethylene
-**ŇŽKL**- polypropylene
-**ŇŽKR**- polycarbonate
-**ŇŽKŘ**- fiberglass
-**ŇŽKF**- linoleum
-**ŇŽKŢ**- polyurethane
-**ŇŽKÇ**- plexiglass (polymethyl methacrylate)
-**ŇŽKH**- epoxy (i.e., polyepoxide)
-**NTV**- latex (synthetic)
-**NZX**- polymer
-**NZXW**- cyanoacrylate
-**NZXL**- polyfluoroalkyl substance (PFAS)

-**PNY**-	cork	-**SNW**-	vinyl (i.e., polyvinyl chloride)
-**NDG**-	resin	-**SNY**-	nylon
-**NZD**-	brine	-**ḌGW**-	paper
-**NZB**-	shellac	-**ḌGY**-	cardboard
-**BGV**-	grease (synthetic lubricant)	-**VNW**-	leather
-**BGŘ**-	non-greasy lubricant	-**VNY**-	cloth/fabric/textile
-**TPY**-	rubber	-**LTN**-	ink

Roots relating to common rocks, minerals, rock-forms, gemstones, and non-mineral materials treated as gemstones:

-**GGŘ**-	mineral	-**GVY**-	diamond
-**GGW**-	stone (material substance)	-**GVM**-	sapphire
-**GGY**-	marble	-**GVN**-	ruby
-**GGL**-	limestone	-**GVW**-	emerald
-**GGR**-	chalk	-**GVL**-	opal
-**GDŘ**-	granite	-**GVG**-	crystal
-**GBŘ**-	basalt	-**STḶ**-	jade
-GBW	serpentine	-**ŘTḶ**-	topaz
-GBY-	syenite	-**ŇTḶ**-	aquamarine
-**GḌL**-	quartz	-**FTḶ**-	amethyst
-**GḌR**-	quartzite	-**ḌTḶ**-	alexandrite
-**GNW**-	slate	-**LTḶ**-	turquoise
-**GNY**-	coal	-**MTḶ**-	spinel
-**GFL**-	gneiss	-**RTḶ**-	malachite
-**GFR**-	calcite	-**PTḶ**-	garnet
-**GFŘ**-	gypsum (calcium sulfate dihydrate)	-**KTḶ**-	zircon
-**GVŘ**-	obsidian	-**JTḶ**-	lapis lazuli
-**GZF**-	feldspar	-**GTḶ**-	ammolite
-**GŽG**-	pyroxene	-**ZTḶ**-	agate
-**GFW**-	mica	-**CTḶ**-	alabaster
-**GFY**-	amphibole		
-**GZL**-	schist	-**ṬDW**-	amber
-**GZR**-	shale	-**ṬDY**-	jet
-**GZM**-	tuff / volcanic tuff	-**ṬDL**-	pearl

-**GZN**- pumice
-**GŢW**- igneous rock
-**GŢY**- sedimentary rock
-**GŢL**- metamorphic rock
-**ŢDŘ**- petrified wood

7.0 NATURAL PHENOMENA (BIOLOGICAL/ZOOLOGICAL)

-**LCP**- **'CELLULAR ORGANELLE / PLASMA MEMBRANE / CYTOPLASM'**		
BSC	**STEM 1:** (to be) an organelle within a biological cell	**STEM 2:** (to be) the plasma membrane / cell membrane of a biological cell
CTE	(to be) the physical body of an organelle within a biological cell	
CSV	(to be) the physical operation of an organelle within a biological cell	**STEM 3:** (to be) the cytoplasm (cytosol) within a biological cell
OBJ	(to be) function/purpose of an organelle within a biological cell	

The following stems follow the same Specification pattern as the above root:

-**PPW**- 1. cellular nucleus 2. nucleolus 3. chromatin
-**PPY**- 1. nuclear envelope 2. nuclear pore 3. endoplasmic reticulum
-**PPL**- 1. ribosome 2. centrosome 3. peroxisome
-**PPR**- 1. mitochondrion 2. lysosome 3. microtubule
-**PPŘ**- 1. Golgi apparatus 2. Golgi vesicle 3. secretory vesicle
-**PPH**- 1. microfilament 2. intermediate filament 3. vacuole